Rituximab around solid-tumour CAR-T

B-cell depletion to limit anti-CAR (anti-idiotype) antibodies and permit repeat dosing: schedules, evidence class, and administration

Literature synthesis of 252 curated records (1988–2026)

No published solid-tumour CAR-T trial has shown that rituximab prevents anti-CAR antibodies or improves CAR-T persistence. The two CAR-T-specific schedules quoted here are investigational trial protocols, not standards of care, and the administration guidance is transferred from the prescribing information and from other indications. Nothing here is a treatment recommendation for an individual patient.

Rituximab / B-cell-depletion dosing to avoid anti-CAR antibodies in solid-tumour CAR-T

Scope: humoral anti-CAR immunogenicity (anti-idiotype / anti-drug antibodies, ADA) in CAR-T therapy of **solid tumours**. Lymphoma, leukaemia and myeloma CAR-T are used only where they supply mechanism or assay methodology, and are explicitly flagged as such. Rituximab used as a CAR **kill switch** or as anti-tumour/target-sensitising therapy is catalogued separately as a confusable, not as evidence.

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1. Bottom line

1. **No rituximab schedule for this indication has been published, but two active trials specify one and their results are not out yet.** Targeted PubMed searching (~3,100 records, `notes/search_strategy.md`) found no publication; ClinicalTrials.gov does (full quotes and URLs in `notes/registry_and_unpublished_evidence.md`): - **Stanford, GD2-CAR T, H3K27M+ diffuse midline glioma, NCT04196413 Arm D** (recruiting): rituximab **750 mg/m²/day IV on days −6 and −5 for the first round, then 750 mg/m² on day −5 for subsequent rounds**; cyclophosphamide 500 mg/m² + fludarabine 30 mg/m² on days −4/−3/−2; GD2-CAR T ICV 10–50 × 10^6 cells on day 0, repeated with re-lymphodepletion. Note this is **2× the conventional 375 mg/m² dose**, and it is repeated per cycle rather than given as a one-off induction. Arms A–C use no rituximab. - **Penn, CART-EGFR-IL13Rα2 (murine EGFR806 scFv + humanised IL13Rα2 scFv), newly diagnosed GBM, NCT06973096 Cohort B**: rituximab **375 mg/m²/day × 1 day** with fludarabine 30 mg/m² and cyclophosphamide 300 mg/m² × 3 days, before each **q6-week** repeat ICV cycle. Cohort A (single dose, no repeat) gets no lymphodepletion and no rituximab. Neither has reported immunogenicity or efficacy data. Any efficacy claim for rituximab in this setting is still extrapolation.
2. **The rationale is now documented, in the same trial that added rituximab.** Chen et al. (medRxiv 2026, PMID 42465905, `fulltext/chen_2026_medrxiv_anticar_dmg.pdf`) show in NCT04196413 that GD2-CAR T therapy induces CD4+/CD8+ anti-CAR T-cell reactivity against murine scFv and junctional epitopes plus circulating **human anti-CAR antibodies (HACAs)** that bind CAR-expressing cells and inhibit CAR-mediated killing; HACA appearance tracked with progression, and HACA level correlated inversely with CAR-T persistence. They state Arm-A outcomes will be compared with "patients currently being enrolled on arms delivering an intensified lymphodepletion regimen designed to reduce or eliminate anti-CAR immune responses" — Arm D. The January 2026 CIRM board presentation (`fulltext/mackall_2026_cirm_board_gd2cart.pdf`) tabulates the arms — anti-CAR immunity in **all** Arm-B patients (no lymphodepletion) and early, versus **many but late** in Arm A, with Arm C (sequential lymphodepletion) showing "lower levels but still present" — and concludes the target product profile "requires sequential intracerebral infusions and sequential lymphodepleting chemotherapy, **likely with rituximab (mirror Arm D)**".
3. **Same preprint gives the reason B-cell-malignancy CAR-T tells you nothing here**: those patients are already immunosuppressed and CD19/BCMA products self-deplete B and plasma cells, "limiting induction of anti-CAR antibody responses". Solid-

tumour patients have intact humoral immunity — e.g. **95% of recipients** of the CLDN18.2 CAR satri-cel in gastric/GEJ cancer developed anti-CAR antibodies (Lancet 2025, PMID 40460847).

4. **The one in-vivo test of anti-CD20 before a CAR-T product failed to prevent anti-CAR antibodies.** Pampusch 2023 (PMID 36825014, non-human primate, SIV — not a tumour) gave **anti-CD20 7 mg/kg IV as a single dose 7 days before CAR-T infusion**. B cells were fully depleted in blood but only partially in lymph nodes, and **all three CAR-T-treated animals still had anti-CAR IgG at day 56**; the authors attribute the failure to incomplete depletion of lymphoid follicles. Pre-treatment was also associated with prolonged IL-6 elevation and faster CAR-T disappearance. This is the closest thing to direct evidence and it is negative for a single low pre-dose.
5. **The problem being solved is real and solid-tumour-specific.** Anti-idiotype/anti-CAR humoral responses have neutralised or curtailed solid-tumour CAR-T in: CAIX CAR-T in renal cell carcinoma (PMID 20889925, 23423337), FRα CAR-T in ovarian cancer (17062687), TAG-72 CAR-T in colorectal cancer (28344808), mesothelin CAR-T where human anti-chimeric antibodies appeared in 8/14 patients (31420241), and most severely mRNA mesothelin CAR-T, where intermittent repeat dosing of a murine scFv produced **anaphylaxis and cardiac arrest** via IgE (24777247).
6. **The field's own B-cell-depletion attempt in a solid tumour used a CAR, not rituximab — and was negative**: in metastatic pancreatic cancer a CD19 CAR was co-infused with a mesothelin CAR to deplete B cells; B cells became undetectable by 7–10 days and stayed so for ≥28 days, but CART-Meso persistence did not improve at the dose tested (n = 3, PMID 32730744).
7. **If a rituximab-containing regimen is to be designed, the transferable schedules come from immune tolerance induction (ITI) in enzyme replacement therapy, AAV gene therapy and haemophilia** (§3), and from one solid-tumour precedent using cyclophosphamide rather than rituximab (§4). All are indirect.
8. **Design-side mitigation is better evidenced than immunosuppression**: humanised/fully human binders, linker choice, and ADA monitoring before redosing (§6).

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2. Direct evidence table (solid tumour / CAR product)

Setting	Product & dosing	ADA finding	B-cell-directed prophylaxis?	PMID
mRCC	CAIX CAR-T, repeated IV infusions ± IL-2	anti-idiotype antibodies **neutralising CAR function** in most patients; also anti-vector immunity	none	20889925, 23423337, 16648493
Ovarian	FRα CAR-T, IV ± IL-2	inhibitory serum factor in **3/6 patients tested**, blocking T-cell responses to FR+ tumour; no persistence	none	17062687
Colorectal	first-generation TAG-72 CAR-T (humanised CC49),	induced **interfering antibody to the CC49 binding domain** (seen as an artefactual fall in serum	none	28344808

Setting	Product & dosing	ADA finding	B-cell-directed prophylaxis?	PMID
	IV or hepatic artery, up to 1e10 cells	TAG-72); persistence ≤14 weeks; authors recommend fully human constructs		
Mesothelioma / pancreas / ovary	**mRNA** mesothelin CAR-T, intermittent repeat infusions	IgE anti-murine-scFv → **anaphylaxis, cardiac arrest on 3rd infusion**	none	24777247, 24579088
Mesothelioma / ovarian / pancreas	lentiviral SS1 (murine scFv) mesothelin CAR-T, single infusion ± Cy	**HACA in 8/14 patients**; fully human CAR taken forward	none	31420241
Pancreas (n = 3)	mesothelin CAR-T **co-infused with CD19 CAR-T**	B cells undetectable 7–10 d and ≥28 d, but **CART-Meso persistence not improved**	yes — CAR-mediated, not rituximab; **negative**	32730744
NHP, SIV (not tumour)	CD4-MBL-CAR/ CXCR5 T cells	anti-CAR IgG in **all** animals, ADCC-competent	—	36582226
NHP, SIV (not tumour)	same CAR + **anti-CD20 7 mg/kg IV, day −7, single dose**	**anti-CAR IgG still developed in all animals**; partial LN depletion only	yes — **failed**	36825014
Gastric / GEJ	satri-cel (CLDN18.2) CAR-T, up to 3 infusions, phase 2 randomised	**anti-CAR antibodies in 95% of recipients**	none	40460847
H3K27M+ DMG	GD2-CAR T (murine 14g2a scFv), IV then sequential ICV, NCT04196413	anti-CAR CD4/CD8 responses + **HACAs that inhibit CAR killing**; HACA inversely correlated with persistence, temporally linked to progression	**Arm D adds rituximab 750 mg/m² d−6/−5 then d−5 per round**; no results yet	42465905 (preprint)
Newly diagnosed GBM	CART-EGFR-IL13Rα2 (murine EGFR806 scFv), q6-week repeat ICV cycles	not yet reported	**rituximab 375 mg/m² ×1 + Flu/ Cy per cycle**, Cohort B only; no results yet	NCT06973096

Solid-tumour trials that depend on repeat dosing — and therefore carry the ADA risk this question is about — are indexed under `C_solid_redosing_trials` (HER2 sarcoma up to 12 infusions, 25800760; repeated intracerebroventricular GD2 / HER2 / B7-H3 dosing, 35130560, 39537919, 34253928, 39775044; repeated intracavitary IL13Rα2, 28029927; hepatic-artery and intraperitoneal CEA CAR-T, 32843493, 31155611, 41760800).

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3. Dosing schedules that do exist, and what they were given for

All extracted verbatim from the sources in `fulltext/` (see `notes/dosing_schedules_extracted.md`). **Evidence class**: none of these were given to prevent anti-CAR antibodies.

Regimen	Timing relative to antigen	Outcome	Setting (evidence class)	PMID
Rituximab **375 mg/m² IV weekly × 4** + methotrexate **0.4 mg/kg × 3 cycles** (with first 3 doses of antigen) + IVIG **500 mg/kg monthly** — "5-week short-course ITI"	started **at or before the first antigen exposure**	prevented/blunted anti-drug antibody; sustained tolerance	CRIM-negative infantile Pompe ERT (prophylactic ITI in humans)	32849613, 22237443, 28814660, 23825616, 26167453
Rituximab **375 mg/m² IV weekly × 3** + methylprednisolone 10 mg/kg, then **sirolimus daily**, **rituximab every 12 weeks**, IVIG monthly (IgG trough 700–1000 mg/dL)	rituximab started **~20 days before** first antigen exposure	**non-responsiveness to both capsid and transgene**, redosing possible	AAV1-GAA gene therapy, single human subject	25541616
Rituximab **20 mg/kg IV (≈375 mg/m²)** + cyclosporine A	after antibody had formed (rescue)	eradicated anti-FIX inhibitors in some animals	NHP AAV8-FIX	22565846
Anti-CD20 mAb **before vector**, ± rapamycin / anti-BAFF / anti-CD19+CD22	**before** first exposure	prevented neutralising antibody, permitted re-administration	mouse/NHP AAV	40057826, 38440160, 36994804, 32670285
Anti-CD20 **+ rapamycin** (4 mg/kg 3×/week) with antigen	concurrent	tolerance where anti-CD20 alone was insufficient	haemophilia A inhibitors (human 2-patient + preclinical)	31985872, 27683758, 32670285
Rituximab 375 mg/m² weekly × 4 **+ bortezomib** 1.3 mg/m² days 1/4/8/11 ± MTX ± IVIG	after high titres established (rescue)	titres fell only when the **plasma-cell arm** was added	Pompe with entrenched ADA	23060045, 27493997
Cyclophosphamide ≥ 4,000 mg/m²	before, and murine protein started **< 90 days after**	HAMA-positivity reduced to 2/76 vs 17/27 if started > 120 days	**neuroblastoma (solid tumour), murine 3F8 antibody**	16421906
Cyclosporin A; deoxyspergualin	around repeated murine antibody dosing	suppressed HAMA, permitted repeat dosing	solid-tumour radioimmunotherapy (historical)	3265333, 8306252, 7606728, 9815950
Standard rituximab **375 mg/m² weekly × 4**; also 1 × 375 mg/m², 2 × 375 mg/m² d0/d7, 1000 mg d0/d14, 500 mg, 100 mg	—	dose determines **duration** of depletion (median B-cell return 2.5 vs 5.0 vs 6.6 months for 100 mg/m², 375 mg/m², 2 × 375 mg/m²)	pharmacology/autoimmunity	9310469, 30109447, 24609057, 40374120, 38247568, 38595918

4. Timing and depth principles that any schedule must respect

- **Prophylaxis works, rescue mostly does not.** Anti-CD20 ablates a de novo antibody response but spares pre-established immunity (35015688); it blunts primary and secondary responses to a true neoantigen (bacteriophage φX174 in humans, 15636611; hapten in primates, 11161973). So rituximab must precede the **first** CAR-T infusion — HAMA-type primary responses appear ~2 weeks after exposure (1711007).
- **An established anti-CAR titre is made by CD20-negative long-lived plasma cells** and will not be reversed by rituximab: plasma cells and their antibody survive years of complete B-cell aplasia after CD19 CAR-T (27166358), and bortezomib only delays rather than eradicates inhibitor-producing cells (21251202). Rescue requires a plasma-cell agent (bortezomib), IgG cleavage (imlifidase, 32483358), or plasmapheresis (38327805, 40705999).
- **Blood depletion ≠ follicular depletion.** This is why the one CAR-T experiment failed (36825014): lymph-node and germinal-centre B cells persist after standard dosing (30649469, 14551029, 15838377, 33384685, 42162633), and higher NHP doses (20–50 mg/kg, often ×3 weekly) are needed for nodal depletion. Type-2 anti-CD20 (obinutuzumab) depletes tissue compartments more deeply (31257724).
- **Inflammation confers resistance to anti-CD20-mediated depletion** (27265023) — directly relevant to a patient with bulky tumour, CRS, or IL-6 elevation.
- **Anti-CD20 alone leaves T-cell help intact**; the regimens that actually induced tolerance paired B-cell depletion with an mTOR inhibitor or methotrexate (32670285, 31985872, 33205401, 28562666).
- **IVIG timing matters**: IVIG given too close to rituximab can block B-cell depletion and abolish ITI efficacy (32681978).

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5. CAR-T-specific hazards of adding rituximab

- **Do not give rituximab to a patient whose CAR-T product carries a CD20/RQR8 safety switch** — rituximab is the agent used to delete those cells (24970931, 19734426, 37086278, 33526841). This is the single most common source of confusion in this literature and is catalogued in `I_confusables_do_not_confuse`.
- Cy/Flu lymphodepletion already reduces B cells; regimen intensity varies (41183747), so the marginal benefit and the marginal toxicity of rituximab must be judged against that baseline.
- Rituximab plus corticosteroids around CAR-T is associated with **late cytopenias** (35842124).
- Prolonged B-cell depletion causes hypogammaglobulinaemia and impaired vaccine responses, requiring IgG monitoring/replacement (36372267, 41942445, 36906276, 34702753).
- B-cell depletion can impair vaccine-induced **CD8 T-cell** responses in a type-I-interferon-dependent manner (34226189) — a theoretical risk of blunting the CAR-T response itself.

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6. Better-evidenced alternatives to immunosuppression

1. **Remove the immunogen**: humanised or fully human binders eliminated the ADA barrier to retreatment in the settings where it was formally studied (33512480, 26101914, 22127019, 41881100); framework engineering did the same for murine antibodies (14522015). The scFv **linker** itself is a dominant epitope (41253492) — a cheap design fix.
2. **Monitor before redosing.** Validated flow-cytometric and cell-based anti-CAR ADA assays exist (42033387, 31678267, 37983584, 41518483); the OKT3-era paradigm of screening before re-exposure prevented anaphylaxis (8497882, 8090628).
3. **Regional/locoregional repeat dosing** (intracerebroventricular, intrapleural, intraperitoneal, intra-arterial) is the delivery strategy actually being used to redose solid-tumour CAR-T (35130560, 39537919, 34253928, 39775044, 34266984, 32843493) — how much systemic humoral immunity constrains it is, to our reading, not yet reported.
4. **Cyclophosphamide** is the only agent with solid-tumour human data for suppressing an anti-murine-protein humoral response (16421906), and it is already part of standard lymphodepletion.

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7. If a schedule must be proposed

Preferred: copy a schedule that is actually being tested in a solid-tumour CAR-T protocol rather than porting one from ERT/AAV. The two options, with no outcome data yet (`notes/registry_and_unpublished_evidence.md`):

	Stanford NCT04196413 Arm D	Penn NCT06973096 Cohort B
Rituximab	750 mg/m²/day IV, days −6 **and** −5 first round; 750 mg/m² day −5 each later round	375 mg/m²/day × 1 day, each cycle
Chemo	Cy 500 + Flu 30 mg/m², days −4/−3/−2	Cy 300 + Flu 30 mg/m² × 3 days
CAR-T	ICV 10–50 × 10^6, day 0, repeated	ICV, q6 weeks
Logic	double the standard dose, before **every** round	standard dose, before **every** round

Both re-dose rituximab with each CAR-T cycle rather than giving a single induction — consistent with §4: blood depletion is easy, follicular/germinal-centre depletion is not, and repeat CAR-T exposure keeps re-priming the response.

The generic ITI-derived design below is **hypothesis-generating** and pre-dates the trial schedules above; it has never been tested with a CAR-T product, and the only in-vivo attempt with a single low pre-dose failed (36825014):

- **Precondition, do not react**: rituximab **375 mg/m² IV weekly, starting ≥ 2–4 weeks before the first CAR-T infusion**, for ≥ 2–4 doses (the ITI/AAV templates: 32849613, 25541616) — with the explicit caveat that standard dosing did not clear nodal/follicular B cells in the CAR-T primate experiment.

- **Verify depth, not just blood counts**: high-sensitivity flow for residual B cells (33384685); consider type-2 anti-CD20 for deeper tissue depletion (31257724).
- **Add a T-help arm** (sirolimus or low-dose methotrexate) rather than relying on B-cell depletion alone (32670285, 31985872, 32849613).
- **Do not expect rescue**: once anti-CAR titres exist, add a plasma-cell-directed or IgG-cleaving strategy (23060045, 32483358) rather than more rituximab.
- **Support IgG** with monthly IVIG, timed away from rituximab dosing (32681978, 36372267).
- **Absolute contraindication**: any CD20-based safety/selection switch in the CAR construct (24970931).
- **Monitor** ADA and CAR-T persistence with a validated assay at each redose (42033387, 31678267).

7b. How the drug is given, once a protocol calls for it

Full detail in `notes/administration_protocol.md` (every block labelled LABEL / REGISTRY / LIT / INFERENCE). The points that change what gets written on an order sheet:

- **Timing within the round**: both trial schedules put rituximab **immediately before lymphodepletion** (day −6/−5 or day −5) and repeat it **every** CAR-T round. Neither registry record specifies premedication, infusion rate, or monitoring.
- **Infusion**: IV only, never push/bolus; dilute to 1–4 mg/mL in NaCl 0.9% or D5W. First infusion 50 mg/h (paediatric 0.5 mg/kg/h, max 50 mg/h), +50 mg/h every 30 min to max 400 mg/h; subsequent infusions start at 100 mg/h. The 90-minute rapid schedule is label-sanctioned only from cycle 2, with a steroid-containing regimen, and not if lymphocytes ≥5,000/mm³ (17244675, 16856919).
- **Premedication**: acetaminophen + H1 antihistamine before every infusion (paediatric label: 30–60 min before). Adding an H2 antagonist does not help (42530421). A second-generation H1 avoids sedation (41989666). **The steroid question is the CAR-T-specific one**: steroids reduce IRR (37205922) but inhibit ADCC (38270799) and blunt CAR-T; give one only if the CAR-T protocol says so.
- **Screening**: HBsAg + anti-HBc mandatory — anti-CD20 is high-risk for HBV reactivation (25% with R-CHOP in HBsAg−/anti-HBc+ patients, 19075267), so past or chronic HBV needs antivirals during and ≥12 months after (32716741, 23775967). Baseline CBC and **baseline quantitative IgG** (omitted in 85% of a 4,479-patient cohort, 30646343).
- **Reactions**: stop or slow, treat by severity, resume at ≥50% rate reduction, discontinue for grade 3–4 (label). Anaphylaxis is rechallengeable by desensitisation, serum sickness is not (42073020) — which matters when the drug is needed again next round.
- **Late effects that overlap the CAR-T window**: late-onset neutropenia at median 77–102 days (20827108, 21560117), hypogammaglobulinaemia (23276889), PML at median 5.5 months (19264918). Fold prophylaxis and IgG replacement into the CAR-T programme's plan (42274870, 34923107).
- **Fasting**: **not required for the rituximab infusion** — the label imposes no NPO status, and high-TLS-risk patients are supposed to be aggressively hydrated. Fasting rules belong to any sedation/anaesthesia for reservoir placement or ICV access, per the local anaesthesia policy (28045707, 36629465, 34857683).

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8. What is not known

- No solid-tumour CAR-T trial has reported an ADA-prophylaxis intervention arm; there is no dose-response, no timing comparison, and no persistence or efficacy endpoint for rituximab in this setting.
- Whether locoregional redosing (ICV, intrapleural) is materially affected by systemic anti-CAR antibodies.
- Whether B-cell depletion impairs CAR-T expansion or antitumour function in humans.
- Whether anti-CAR ADA in solid tumours is predominantly IgG (neutralising/ADCC) or IgE (anaphylaxis), which determines whether prophylaxis is a persistence question or a safety question (24777247 vs 20889925).

Extracted dosing schedules (verbatim from the primary sources)

Quotations are from the full texts in `../fulltext/` unless the source is marked (abstract only). Nothing here was given to prevent anti-CAR antibodies in a solid tumour; the indication is stated for each.

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1. The only anti-CD20 dosing given before a CAR-T product

PMID 36825014 — Pampusch 2023, Front Immunol, SIV-infected ART-suppressed rhesus macaques, CD4-MBL-CAR/CXCR5 T cells. Not a tumour.

"Seven days prior to infusion of CAR-T cells, the CD20-depleted animals were treated intravenously with a dose of 7 mg/kg of a rhesus IgG1 recombinant Anti-CD20 (2B8R1) monoclonal antibody"

"We chose to similarly administer a single low dose (7 mg/kg) of anti-CD20 7 days prior to CAR-T cell infusion in order to facilitate a temporary depletion of the B cell follicles."

Outcome — **prophylaxis failed**:

"In the CD20-depleted animals, at 56 DPT, all three of the CAR/CXCR5-treated animals also had detectable levels of anti-CAR IgG antibodies in their serum … This finding suggests that the ability of the animals to develop antibodies to the CAR was not impeded by anti-CD20 treatment, perhaps due to the limited depletion of the lymphoid follicles."

Depletion depth achieved, and the doses the authors say are actually needed in primates:

"In CD20 depleted animals, complete depletion was observed in the blood and partial depletion was observed in the lymph nodes nine days after administration of 7 mg/kg anti-CD20."

"Previous primate studies used anti-CD20 antibody doses of 20-50 mg/kg, usually with multiple administrations, which had prolonged impacts on CD20+ cells in blood and lymphoid tissues."

"Infusion of 1.6 or 6.4 mg/kg rituximab … resulted in over 95% depletion of B cells in peripheral blood for 8 days and variable CD20 depletion in lymph nodes, resulting in 34-77% depletion of lymph node B cells seen 2 and 4 weeks after treatment." / "Three weekly doses of 20 mg/kg rituximab were shown to significantly deplete B cells in the peripheral blood and lymph nodes of rhesus macaques, also resulting in decreased Ki67+ B cells in GCs".

Additional CAR-T-relevant harms: "The potential for an inflammatory cytokine response appears to be enhanced with anti-CD20 antibody treatment and future studies may require CRS control strategies", and the CAR-T cells in depleted animals "abruptly disappeared" after an early proliferative burst.

Companion paper **PMID 36582226** (same CAR, no depletion) established the ADA baseline: "All of the treated animals developed antibodies in their serum that bound to CD4-MBL CAR/CXCR5 T cells and the majority were capable of inducing an ADCC response", with the dominant epitopes in the CD4 domains and the CD28 transmembrane/linker region — i.e. **ADA occurs even with self-protein-based CARs**.

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2. Prophylactic ITI with rituximab, given before first antigen exposure (Pompe ERT)

PMID 32849613 — Desai 2020, Front Immunol, CRIM-negative infantile Pompe, prophylactic short-course ITI:

"The 5-week short-course immune tolerance induction approach included four doses of weekly rituximab (375 mg/m², intravenously), and three cycles of low-dose methotrexate (0.4 mg/kg …)"

"To provide passive immunity during B cell suppression, monthly IVIG at 500 mg/kg was added to the combination therapy during the time of B cell suppression."

"ITI with rituximab (4 weekly doses), methotrexate (3 cycles with first 3 ERT infusions …)" — i.e. the MTX cycles are timed to the **first three antigen exposures**.

PMID 22237443 (Messinger 2012, Genet Med) and **PMID 23825616** (Banugaria 2013) give the same protocol: "a regimen of rituximab (four weekly doses intravenously) and methotrexate (three doses per week for three weeks subcutaneously), with or without monthly IVIG".

PMID 26167453 (Stenger 2015) — sibling comparison showing timing is the variable that matters: "prophylactic ITI with rituximab, methotrexate, and IVIG was initiated at the time of first alglucosidase alfa dose (20 mg/kg every 2 weeks)" and "Patient 2 received a five week course of prophylactic ITI with rituximab, IVIG and methotrexate which was initiated **prior to** first dose of alglucosidase alfa".

PMID 32681978 (abstract only) — IVIG given too close to rituximab can block depletion and abolish ITI efficacy; IVIG must be timed away from the rituximab doses.

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3. Rituximab before AAV exposure — the closest human template for pre-CAR prophylaxis

PMID 25541616 — Corti 2014, Mol Ther Methods Clin Dev, single human subject, rAAV1-CMV-hGAA:

> "At 5.5 months of age, about 20 days before starting ERT, the patient received 375 mg/m² of rituximab and 10 mg/kg of methylprednisolone intravenously (premedication) weekly for 3 weeks."
>
> "After the Rituximab induction doses, the patient received daily oral Sirolimus (0.06–1 mg/m²/day)."
>
> "Once the induction dose of Rituximab was completed, the patient started receiving a monthly dose of 500–1,000 mg/kg of IV immunoglobulin (IVIG) … adjusted in order to maintain a trough serum IgG level of 700–1,000 mg/dl."
>
> "A 3.5 year-old Pompe disease subject … received weekly administration of 375 mg/m² of rituximab along with 10 mg/kg of methylprednisolone for 3 weeks prior to dosing with 5 × 10^{12} vg/kg of rAAV1-CMV-hGAA. The patient then received daily oral administration of 0.06–1 mg/m² of Sirolimus through the study and continued B-cell depletion with **Rituximab every 12 weeks**."

Result: "B-cell ablation with rituximab **prior to** AAV vector exposure results in non-responsiveness to both capsid and transgene, therefore allowing the possibility of repeat administration in the future."

PMID 22565846 — Mingozzi 2012, NHP, anti-FIX inhibitor eradication (rescue, not prophylaxis): "Rituximab … was given intravenously at a dose of **20 mg/kg (equivalent to ~375 mg/m²)**" combined with cyclosporine A.

PMID 40057826 — Doshi 2025, mouse AAV8: "anti-CD20 initiated **before** AAV8 exposure with or without concurrent immunosuppression prevented AAV8 NAb development and permitted transgene expression with systemic AAV8 vector re-administration"; regimen "anti-CD19 (150 µg weekly) and anti-CD22 (150 µg weekly) with either anti-B220 (150 µg weekly) or **anti-CD20 (250 µg every 21 days)**".

PMID 38440160 — Rana 2024: transient protocol combining **anti-CD20 (B-cell depletion) + anti-BAFF** (to slow repopulation) ± rapamycin "100 µg (~4 mg/kg) … by oral gavage 3×/week for 5 weeks"; "A single α-CD20 treatment depleted >99% of circulating B cells by day 2", and the combination "allows for efficient re-administration following immune cell repopulation" at 12 weeks.

PMID 22704060 (negative control) — transient intensive immunosuppression (MMF-based) failed to improve repeat AAV5 liver gene transfer in macaques.

PMID 40705999 (abstract only) — clinical de-titring before AAV9 gene therapy: "11 plasma exchanges and 2 doses of rituximab" lowered anti-AAV9 titres from 1:800 into the eligibility range; both twins still had acute hypersensitivity reactions during infusion.

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4. Pairing anti-CD20 with an mTOR inhibitor (haemophilia inhibitors)

PMID 31985872 — Doshi 2020, J Thromb Haemost, two paediatric patients with refractory inhibitors:

> "treated with 100 units/kg/d of recombinant FVIII, rapamycin to target trough values of 5–15 ng/mL, and **four weekly doses of 375 mg/m^2 rituximab**"

PMID 32670285 — Biswas 2020, AAV8-coF8 mice: "anti-mCD20 (250 µg) on weeks 0 and 3 or oral rapamycin (4 mg/kg) 3 times per week from weeks 2 through 5 post-vector administration"; the combination gave a consistently larger reduction in inhibitor titres (3.8–8.2-fold) than anti-CD20 alone (0.43–2.2-fold).

PMID 27683758 — Biswas 2017: "Naïve BALB/c mice were injected IV with two doses of 250 µg anti-CD20 … three weeks apart", followed by "rapamycin (4 mg/kg, 3×/week for 4 weeks)".

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5. Rescue of an established titre needs a plasma-cell arm

PMID 23060045 — Banugaria 2013, Genet Med, Pompe with titres up to 1:204,800:

> "Cyclophosphamide (250 mg/m^2 i.v.) monotherapy was administered at weeks 86 and 92 of ERT, followed by rituximab (375 mg/m^2 i.v.) every week from week 92 to week 95"

> "an ITI protocol … included rituximab (375 mg/m^2 i.v., every week for 4 weeks) and methotrexate (15 mg/m^2 s.c.) every other week." → "Even after his B-lymphocyte CD19 count dropped from 17.9% … to 0.1%, antibody titers did not drop to <1:102,400."

> "Bortezomib was administered twice weekly (1.3 mg/m^2 i.v.) according to a standard dosing regimen (days 1, 4, 8, and 11; equivalent to one cycle)" → titres fell from 1:204,800 to 1:51,200 and eventually 1:3,200.

Consistent with **PMID 27166358** (plasma cells and their antibody survive years of complete B-cell aplasia after CD19 CAR-T) and **PMID 21251202** (bortezomib delays but does not eradicate inhibitor-producing cells).

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6. The solid-tumour precedent: cyclophosphamide, not rituximab

PMID 16421906 (abstract only) — Kushner 2007, neuroblastoma, murine antibody 3F8:

"the incidence of HAMA-positivity was significantly lower if patients received **high-dose cyclophosphamide (HD-Cy, ≥ 4,000 mg/m²)** before 3F8 treatment (P < 0.001). In addition, HAMA-positivity was least likely if 3F8 treatment was initiated **< 90 days post-HD-Cy** (2/76 compared to 3/19 first treated at 90–120 days, and 17/27 first treated at > 120 days)"

"HD-Cy reliably blocks humoral responses to a murine antibody. This capacity to prevent host rejection of foreign or not fully humanized proteins raises the possibility of a broad role for HD-Cy in immunotherapeutic strategies."

Historical companions: cyclosporin A (PMID 3265333, 8306252, 10541370) and deoxyspergualin (PMID 7606728, 9815950) suppressed HAMA to permit repeated murine-protein dosing in solid tumours.

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7. Timing principle: prophylaxis, never rescue

PMID 35015688 (abstract only) — "None of 31 patients who had received anti-CD20 treatment within 6 months prior to vaccination developed blocking antibodies. In contrast, patients who initiated anti-CD20 treatment shortly after achieving a vaccine-induced antibody response tended to retain that response during treatment, suggesting a policy of immunizing prior to treatment whenever possible." (Cohort is lymphoma; used for the **timing principle only**.)

PMID 15636611 (abstract only) — rituximab significantly decreased the primary and secondary human antibody response to the neoantigen bacteriophage φX174.

PMID 11161973 (abstract only) — anti-CD20 infusion inhibited hapten-induced primary **and** memory humoral responses in primates.

PMID 1711007 (abstract only) — kinetics of primary HAMA responses; the window that prophylaxis must precede is roughly the first two weeks after exposure.

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8. Dose → depth → duration of depletion (what a schedule buys you)

- **PMID 9310469** — the original IDEC-C2B8 schedule: 375 mg/m² weekly × 4.
- **PMID 30109447** — children, one dose 100 mg/m² vs one dose 375 mg/m² vs two doses 375 mg/m² (d0, d7): median time to B-cell reconstitution **2.5 vs 5.0 vs 6.6 months**.
- **PMID 24609057** — a single 375 mg/m² dose achieved B-cell depletion < 0.005 cells/µl in 89% by a median of 13 days, with median time to repopulation 9.2 months.
- **PMID 40374120** — 100 mg vs 500 mg vs 1000 mg at weeks 0 and 2: "CD20+ B-cell depletion occurred universally by week 2, with faster reconstitution in [ultralow dose] by 26 weeks".
- **Tissue vs blood**: PMID 30649469 (incomplete lymph-node depletion after standard dosing), 14551029 and 15838377 (NHP blood vs lymphoid tissue dose-dependence), 33384685 (high-sensitivity detection of residual B cells), 42162633 (depth-of-depletion paradox), **27265023 (inflammation causes resistance to anti-CD20-mediated depletion)**.

- **Type-2 anti-CD20**: PMID 31257724 — obinutuzumab achieves deeper tissue depletion than rituximab.
- **Cost**: PMID 36372267 (acquired B-cell deficiency / hypogammaglobulinaemia), 41942445, 36906276, 34702753 (vaccine responses), 35842124 (late cytopenias when rituximab is used around CAR-T), 34226189 (B-cell depletion impairs vaccine-induced CD8 T-cell responses).

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9. Hard contraindication

PMID 24970931 / 19734426 / 37086278 / 33526841 — CD20 and RQR8 are used as **selection/suicide markers** in engineered T cells; rituximab is the agent that deletes the product. Rituximab must never be given as ADA prophylaxis to a patient whose CAR construct contains a CD20-based switch.

Rituximab in solid-tumour CAR-T: trial-registry and not-yet-published evidence

Added after the PubMed-only round. PubMed indexes no published rituximab schedule for preventing anti-CAR antibodies in solid-tumour CAR-T — but two active trials now build rituximab into the conditioning regimen, and one of them has a public dose and timing. Results are unpublished (one preprint, one funder board presentation).

Retrieved 2026-08-06. All quotes verbatim.

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1. Stanford GD2-CAR T for H3K27M+ diffuse midline glioma — NCT04196413, Arm D

The only registry record giving an explicit rituximab dose/schedule around a solid-tumour CAR-T.

Source: https://clinicaltrials.gov/study/NCT04196413 (record last updated 2026-01-23; status recruiting)

Arm D description:

"GD2CART will be administered in escalating doses on Day 0 in hospitalized subjects with either DIPG, spinal DMG, or high risk features. Intracerebroventricularly after administration of conditioning lymphodepletion chemotherapy regimen with rituximab, cyclophosphamide and fludarabine"

Intervention descriptions (Arm D):

Rituximab — "First round: 750 mg/m2 per day IV for days -6 and -5. Subsequent rounds: 750 mg/m2 per day IV for day -5." Cyclophosphamide — "Cyclophosphamide 500 mg/m2 per day IV for days -4, -3, -2" Fludarabine — "Fludarabine 30 mg/m2 per day IV for days -4, -3, -2"

So, per cycle:

Day	Arm D
−6	rituximab 750 mg/m² IV (**first round only**)
−5	rituximab 750 mg/m² IV (every round)
−4, −3, −2	cyclophosphamide 500 mg/m² + fludarabine 30 mg/m² IV daily

Day	Arm D
0	GD2-CAR T ICV, 10–50 × 10^6 cells
—	repeated with re-lymphodepletion each round

Note the dose: **750 mg/m², i.e. 2× the conventional 375 mg/m² lymphoma/autoimmune dose**, given 5–6 days pre-infusion, and repeated with each dosing cycle rather than as a one-off induction. Arms A–C of the same trial use no rituximab (Arm A single lymphodepletion + IV then ICV; Arm B ICV with no lymphodepletion; Arm C sequential lymphodepletion + sequential ICV).

2. Why Arm D exists: anti-CAR immunity as the resistance mechanism (preprint, not peer reviewed)

Chen Y, Reynolds K, … Ramakrishna S, Mackall CL. Anti-CAR Immunity Drives Acquired Therapeutic Resistance to GD2-CAR T Cell Therapy in Diffuse Midline Glioma. medRxiv, posted 2026-07-09. PMID 42465905 · PMC13370534 · doi 10.64898/2026.06.25.26356492 · https://www.medrxiv.org/content/10.64898/2026.06.25.26356492v1 Local copy: `fulltext/chen_2026_medrxiv_anticar_dmg.pdf`

Abstract:

> "peripheral blood CD4+ and CD8+ T cells manifested anti-CAR immune reactivity targeting epitopes enriched within murine-derived and engineered junctional regions of the CAR construct. This was associated with appearance of circulating Human Anti-CAR Antibodies (HACAs) that bound cells expressing the GD2-CAR, as well as clonal expansion of CSF B cells which produced HACA which impeded the cytotoxic activity of GD2-CAR T cells. In several cases, appearance of circulating HACA temporally correlated with disease progression … levels of circulating HACA inversely correlated with circulating CAR T cell persistence."

Discussion — the forward-looking statement that Arm D is the test of the hypothesis:

> "We anticipate that additional evidence to confirm the role of anti-CAR responses in therapeutic resistance will be gleaned by comparing outcomes from patients treated on Arm A to patients currently being enrolled on arms delivering an intensified lymphodepletion regimen designed to reduce or eliminate anti-CAR immune responses."

The preprint does not itself name rituximab; the drug and schedule come from the registry record (§1) and the funder presentation (§3).

Why solid tumours specifically (same discussion):

> "Most patients who have received investigational and commercial CAR T cells to date have been treated for refractory B cell and plasma cell malignancies and therefore were highly immunosuppressed at the time CAR T cells were administered. Further, all commercial CAR T cells today target CD19 or BCMA and thus eradicate B and plasma cells respectively, limiting induction of anti-CAR antibody responses."

i.e. the reason lymphoma/leukaemia/myeloma CAR-T experience is not transferable is that those products are self-B-cell-depleting. This is the mechanistic justification for the scope exclusion used in this collection.

Corroborating solid-tumour datapoint cited there: in the satri-cel (CLDN18.2) gastric/GEJ phase 2 (CT041-ST-01, Lancet 2025;405:2049–60, PMID 40460847), **95% of CAR-T recipients developed anti-CAR antibodies**.

3. Interim arm-by-arm comparison, including Arm D — CIRM board presentation, January 2026

Mackall CL. GD2-CAR T Cells for Diffuse Midline Gliomas, CIRM Board Meeting, January 2026 (CLIN2-12595). https://www.cirm.ca.gov/wp-content/uploads/2026/01/7b.-20290129-Closer-to-Cures-Presentation.pdf Local copy: `fulltext/mackall_2026_cirm_board_gd2cart.pdf`

Slide table (transcribed):

Arm	Regimen	Outcome	CAR persistence	Anti-CAR immunity
A (n=13)	lymphodepletion ×1, IV ×1, sequential ICV	median OS 20.6 mo, sustained benefit; dose-limiting CRS	greater persistence correlates with better outcomes	"Many patients, occur late"
B (n=18)	no lymphodepletion, sequential ICV	median OS 14.8 mo, transient benefit	"Very poor CAR persistence"	"All patients, occur early"
C (n=11)	sequential lymphodepletion, sequential ICV	median OS not reached	"Better CAR persistence"	"Preliminary data shows lower levels but still present"
D (now enrolling)	**sequential lymphodepletion + rituximab**, sequential ICV	too early	pending	pending

Conclusion slide:

> "We have learned that the target product profile requires sequential intracerebral infusions and sequential lymphodepleting chemotherapy, likely with rituximab (mirror Arm D)."

This is the strongest statement available that a rituximab-containing regimen is the intended solid-tumour CAR-T schedule — but note it is a funder board slide, arms are non-randomised and sequentially enrolled, and no Arm D immunogenicity or efficacy data exist yet.

Related conference record (data overlapping the preprint): Chen Y et al., Longitudinal single-cell atlas of GD2-CAR T cell therapy in H3K27M-mutant diffuse midline glioma identifies humoral and cellular anti-CAR immunity, AACR 2026.

4. Penn CART-EGFR-IL13Rα2 in newly diagnosed GBM — NCT06973096, Cohort B

Source: https://clinicaltrials.gov/study/NCT06973096 (phase 1, started 2025-07-18, active not recruiting)

> Cohort B (Repeat Cycles Following Lymphodepletion): "CART-EGFR-IL13Ra2 cells will be administered via intracerebroventricular delivery in q6 week cycles of treatment, following lymphodepletion with fludarabine, cyclophosphamide, and rituximab."

Interventions:

> "Rituximab or Rituximab biosimilar — 375 mg/m2/day x 1 day" "Fludarabine: 30mg/m2/day x 3 days; Cyclophosphamide: 300mg/m2/day x 3 days"

Product (relevant to why ADA is expected):

> "autologous T cells transduced with a bicistronic lentiviral vector containing a murine scFv targeting EGFR epitope 806 and a humanized scFv targeting IL13Ra2"

So Penn's schedule is **rituximab 375 mg/m² × 1 dose per cycle**, with Flu 30 / Cy 300 mg/m² × 3 days, before each q6-week ICV redose — half the Stanford dose, single administration, but repeated per cycle. Cohort A (single fixed dose, no repeat) receives no lymphodepletion and no rituximab, so the rituximab is tied specifically to the repeat-dosing cohort. The registry record does not state the rationale in words.

Predecessor trials, for contrast — no rituximab, single ICV dose, no lymphodepletion: - NCT05168423 (recurrent GBM, phase 1; published Bagley et al., Nat Med 2025, doi 10.1038/s41591-025-03745-0; ASCO 2025 abstract JCO 43:16_suppl 102; ASCO 2026 OS update JCO 44:16_suppl 2013). - NCT07209241 (phase 1b, recurrent GBM, three dosing approaches incl. repeat dosing) — registry record lists no rituximab as of retrieval.

5. What is not there

- **No B7-H3 CAR-T trial at the University of Pennsylvania.** Searched ClinicalTrials.gov by lead sponsor (53 Penn CAR-T studies, none B7-H3/CD276), by intervention (20 B7-H3 CAR-T studies: St. Jude, Stanford, Seattle Children's, MacroGenics, PersonGen, Zhejiang, Tiantan, Children's National, Tcelltech, BrainChild Bio), and by Philadelphia location. The only B7-H3 CAR-T with a Philadelphia site is **BrainChild Bio's BCB-276 (NCT07680439, "Illuminate")**, a phase 2 DIPG study with CHOP as a site: up to 15 intraventricular doses q14 days, **no lymphodepletion and no rituximab** in the record. Majzner's B7-H3.CD28Z.CART trials (NCT07358260, NCT07390539) are Dana-Farber/ Boston Children's, Flu/Cy only. Seattle's BrainChild-03 (NCT04185038; PMID 39775044, 36259971) gives repeated ICV B7-H3 CAR-T **without** lymphodepletion and reports no rituximab.
- **No published Arm D or Cohort B results** for either rituximab-containing regimen.
- No randomised comparison of rituximab vs no rituximab in any CAR-T setting for ADA prevention.

Giving rituximab around CAR-T: dose, timing, premedication, monitoring, and bedside practicalities

Companion to `REPORT.md`. `REPORT.md` answers whether to give rituximab for anti-CAR immunity (answer: unproven, two trials are testing it). This note answers how the drug is actually given once a protocol calls for it.

Evidence status of each block below is labelled. Nothing here is a substitute for the treating protocol: the two solid-tumour CAR-T regimens are investigational, and no anti-CAR-prophylaxis outcome data exist for either.

Label	Meaning
[LABEL]	US prescribing information, Rituxan IV, label effective 2025-01-06 (openFDA `drug/label`). Established.
[REGISTRY]	Verbatim from the ClinicalTrials.gov record. Investigational, no outcome data.
[LIT]	Peer-reviewed evidence, cited by PMID; strength varies and is stated.
[INFERENCE]	Our reasoning, not a sourced recommendation.

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1. Dose and timing

1.1 The two solid-tumour CAR-T schedules [REGISTRY]

	Stanford NCT04196413 Arm D (GD2 CAR-T, H3K27M+ DMG)	Penn NCT06973096 Cohort B (CART-EGFR-IL13Rα2, ndGBM)
Rituximab	**750 mg/m²/day IV on day −6 and day −5** for the first round; **750 mg/m²/day IV on day −5** for each subsequent round	**375 mg/m²/day IV × 1 day** (rituximab or biosimilar), before each cycle
Lymphodepletion	cyclophosphamide 500 + fludarabine 30 mg/m²/day, days −4, −3, −2	cyclophosphamide 300 + fludarabine 30 mg/m²/day × 3 days
CAR-T	intracerebroventricular, day 0, repeated rounds	intracerebroventricular, q6-week cycles
Design logic	~2× the standard single dose, split over 2 days, **1 day before lymphodepletion starts**, repeated every round	standard single dose, repeated every cycle

Both put rituximab **immediately before lymphodepletion, i.e. 5–6 days before the CAR-T infusion**, and repeat it with every CAR-T round rather than giving a one-off induction. Neither registry record specifies premedication, infusion rate, or monitoring — those come from the treating protocol and the label.

1.2 Reference doses that anchor those numbers [LABEL] / [LIT]

- 375 mg/m² IV is the label dose for B-cell NHL (adult and paediatric ≥6 months) and the GPA/MPA induction dose; it comes from the pivotal 375 mg/m² weekly × 4 trial (9310469) **[LABEL]**.

- CLL uses 375 mg/m² cycle 1 then **500 mg/m² cycles 2–6**; RA and PV use **flat 1,000 mg × 2, 2 weeks apart** — so a >375 mg/m² dose is not itself unusual **[LABEL]**.
- Dose sets the **duration** of depletion more than the depth of blood depletion: median B-cell return ~2.5 / 5.0 / 6.6 months after 100 mg/m², 375 mg/m², and 2 × 375 mg/m² (see `REPORT.md` §3).
- Exposure is not fixed by BSA: clearance rises with proteinuria and with anti-rituximab antibodies (36420256), and anti-drug antibodies shorten depletion (58.5 vs 163 days in children, 42481732; 42340364) **[LIT]**.
- Subcutaneous rituximab (1,400 mg with rHuPH20) is bioequivalent-by-design in lymphoma (24002601, 30744432) but **is label-restricted to patients who have already tolerated a full IV dose**, and neither CAR-T protocol uses it **[LABEL]**.
- Biosimilars are explicitly permitted in NCT06973096; switching studies show no consistent immunogenicity or efficacy penalty (29500555) **[LIT]**.

1.3 Interaction with the CAR-T timeline [INFERENCE]

- Rituximab is given **before** Cy/Flu in both protocols. Cy/Flu itself depletes B cells, so the marginal benefit and marginal toxicity of rituximab is judged against that baseline (see `REPORT.md` §5).
- Fludarabine exposure drives CAR-T expansion and outcome, and is under-dosed in young children by fixed mg/m² dosing (40148484, 38191740, 41471106, 40609087) — adding rituximab does not change that, but it does add a day to the pre-infusion window in which the lymphodepletion schedule must not slip.
- The prophylaxis→rescue asymmetry (`REPORT.md` §4) means a **missed or delayed** pre-round rituximab dose cannot be made up after the CAR-T infusion.

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2. Preparation and infusion [LABEL]

- **IV infusion only. Never IV push or bolus.** Administer only where severe infusion reactions can be managed.
- Dilute to **1–4 mg/mL** in 0.9% NaCl or 5% dextrose. Do not mix with other drugs. Diluted solution: 24 h at 2–8 °C (plus a further 24 h at room temperature shown stable; refrigerate because there is no preservative). Inspect — should be clear and colourless.
- **First infusion (adult):** start **50 mg/h**; if no toxicity, increase by 50 mg/h every 30 min to a maximum of **400 mg/h**.
- **First infusion (paediatric mature B-cell NHL/B-AL):** start **0.5 mg/kg/h (max 50 mg/h)**; increase by 0.5 mg/kg/h every 30 min to max 400 mg/h.
- **Subsequent infusions (adult):** start **100 mg/h**; increase by 100 mg/h every 30 min to max 400 mg/h. Paediatric: start 1 mg/kg/h (max 50 mg/h), increase by 1 mg/kg/h every 30 min.
- **90-minute rapid infusion** (20% of the dose in the first 30 min, remaining 80% over 60 min) is a label option **from cycle 2 onward**, only if no grade 3–4 infusion reaction in cycle 1 and only with a glucocorticoid-containing regimen. **Excluded:** clinically significant cardiovascular disease, or circulating lymphocytes ≥5,000/mm³.

- Rapid-infusion safety data: 90 min in 206 patients with no grade 3–4 reactions (17244675), 319 infusions with and without steroids with no grade 3–4 events (16856919), and 60-min infusions in 105 second-and-later doses (19220420) **[LIT]**.
- **[INFERENCE]** In a repeat-cycle CAR-T protocol, every rituximab dose after the first is a "subsequent infusion", so the faster start rate applies — but the 90-minute option assumes concurrent steroids, which is exactly what §3 says to think twice about here.

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3. Premedication

[LABEL] Premedicate with **acetaminophen/paracetamol and an antihistamine before every infusion**. For paediatric mature B-cell NHL/B-AL the label specifies acetaminophen + an H1 antihistamine (diphenhydramine or equivalent) **30–60 min before** each infusion. A glucocorticoid is label-recommended only for the autoimmune indications (methylprednisolone 100 mg IV or equivalent 30 min pre-dose for RA/GPA/MPA/PV) and, for lymphoma, as the chemotherapy's own steroid when using the 90-minute rate.

[LIT] What the premedication evidence supports:

Component	Evidence
Acetaminophen + H1 antihistamine	Standard of care; label-mandated. Most reactions occur 30–120 min into the first infusion (label; 42530421 observed most IRR at 30–60 min).
Second-generation H1 (e.g. bepotastine) instead of diphenhydramine/hydroxyzine	Randomised phase II: grade ≥2 IRR 32% vs 52% and less drowsiness, not significant at n=40 (41989666). Reasonable when sedation is undesirable.
Adding an **H2** antagonist	No benefit: IRR 38% vs 36.5%, no difference in grade 3–4 (42530421, n=299).
Corticosteroid premedication	Reduces IRR (prednisone pretreatment 15.9% vs 43.2% in DLBCL, 37205922); oral prednisolone 50 mg ≈ IV methylprednisolone 50 mg (41572843, 2,253 infusions). But rapid infusion is safe without steroids (16856919) and steroid necessity is being re-examined (41789935).

Corticosteroids are the one premedication decision that is CAR-T-specific. Steroids inhibit ADCC and are argued to belong at the first dose only for ADCC-dependent mAbs (38270799) **[LIT]**; separately, steroids blunt CAR-T expansion and function and are the treatment for CRS/ICANS rather than a routine co-medication (42274870, `REPORT.md` §5, where rituximab + steroids around CAR-T is linked to late cytopenias). Rituximab here is given days **before** the CAR-T infusion, so a single pre-infusion steroid dose is pharmacodynamically distant from the cells — but this is **[INFERENCE]**, and the decision belongs to the CAR-T protocol, not to the infusion nurse's standing premedication order.

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4. Before the first dose: screening and baseline labs [LABEL] + [LIT]

Test	Why
HBsAg + anti-HBc (add anti-HBs)	Label-mandated before any rituximab dose. Anti-CD20 is a **high-risk** therapy for HBV reactivation: 25% reactivation with R-CHOP in HBsAg−/anti-HBc+ DLBCL vs 0% with CHOP, including a hepatic death (19075267). ASCO: chronic **or past** HBV + anti-CD20 → antiviral prophylaxis during and for **≥12 months after** therapy, with

Test	Why
	HBV-expert co-management (32716741, 25964247, 28219691). Entecavir prophylaxis cut reactivation 17.9%→2.4% in resolved HBV, and an undetectable baseline HBV DNA is **not** protective (23775967). Reactivation has been reported up to 24 months after therapy.
CBC with differential and platelets	Label-mandated pre-first-dose; then before each course (monotherapy) or weekly-to-monthly with chemotherapy, more often if cytopenic. Continue after the last dose until cytopenias resolve.
Baseline quantitative IgG (± IgA/IgM) and B-cell flow	85% of a 4,479-patient cohort had **no** pre-rituximab immunoglobulin measurement, and severe infections rose 17.2%→21.7% (30646343); baseline + serial IgG and B-cell counts are the recommended monitoring (36706910). Also the only way to interpret post-CAR-T hypogammaglobulinaemia (34757064).
Renal function, electrolytes, phosphate, uric acid, LDH; hepatitis C and HIV per CAR-T program	TLS and renal toxicity are label warnings; CAR-T best-practice screening panels (34923107, 31753925).
Anti-CAR ADA baseline, and confirmation of the CAR construct	see §7.

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5. During and immediately after the infusion

[LABEL] Infusion-related reactions are the boxed warning: fatal reactions have occurred within 24 h, **~80% with the first infusion**, severe reactions typically **30–120 min** into it. Manifestations include urticaria, hypotension, angioedema, hypoxia, bronchospasm, pulmonary infiltrates, ARDS, MI, ventricular fibrillation, cardiogenic shock, anaphylactoid events, death.

Management, per label:

1. **Interrupt or slow the infusion** for any infusion-related reaction.
2. Institute medical management as needed — **glucocorticoids, epinephrine, bronchodilators, oxygen**.
3. On symptom resolution, **resume at ≥50% rate reduction** (label also phrases this as one-half the previous rate).
4. **Temporarily or permanently discontinue** depending on severity and interventions required; discontinue for severe (grade 3–4) reactions and for serious/life-threatening arrhythmias.
5. Monitor closely, with cardiac monitoring during and after infusion, in: pre-existing cardiac or pulmonary disease, prior cardiopulmonary reactions, history of arrhythmia/angina, and circulating malignant cells ≥25,000/mm³.

[LIT] Practical additions:

- **Cytokine-release-type reactions and true (IgE) hypersensitivity are clinically indistinguishable at the bedside** but differ in mechanism and in whether rechallenge is safe (20350882, 31447561, 38452439); the distinction is made retrospectively, so treat by severity in the moment.

- **Anaphylaxis is rechallengeable via desensitisation; serum sickness is not.** In 1,534 paediatric infusions: 7 rituximab-induced serum sickness (fever + rash + arthralgia 1–30 days later, raised CRP/ESR, low complement) and 7 anaphylaxis; 5/7 anaphylaxis patients were successfully re-dosed under desensitisation, whereas RISS recurred severely (42073020). Adult desensitisation protocols: 46 procedures in 11 patients without anaphylaxis (42279029); paediatric workup framework in 41937341.
- **[INFERENCE]** In a repeat-cycle CAR-T protocol this matters more than usual: a reaction that closes the door on rituximab closes it on **every future round**, so grade the event carefully and involve allergy/immunology early rather than abandoning the drug.

[LABEL] Also mid-to-late: severe mucocutaneous reactions (including SJS/TEN, onset as early as day 1 → discontinue permanently), TLS within **12–24 h** of the first infusion in high-burden disease (give aggressive IV hydration + anti-hyperuricaemic therapy, correct electrolytes, monitor renal function and fluid balance, dialyse if needed; 21561350), renal toxicity (discontinue for rising creatinine or oliguria), and bowel obstruction/perforation with chemotherapy (evaluate abdominal pain or repeated vomiting).

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6. After the dose: infection prophylaxis, immune reconstitution, vaccines

[LABEL]

- PJP (Pneumocystis) **and herpesvirus** prophylaxis is label-recommended in CLL during treatment and for up to 12 months after; PJP prophylaxis for GPA/MPA during and for ≥6 months after the last dose.
- Serious bacterial, fungal, and new or reactivated viral infection (CMV, HSV, VZV, parvovirus B19, WNV, HBV, HCV) can occur during and after therapy; infections have been reported with hypogammaglobulinaemia persisting >11 months. Withhold rituximab for serious infection; avoid in severe active infection.
- **Live viral vaccines are not recommended before or during treatment**; bring immunisations up to date beforehand where possible.

[LIT]

- Post-CAR-T infection prevention is a structured, risk-adapted programme in its own right — antiviral and PJP prophylaxis, selective antibacterial/antifungal cover during prolonged cytopenia, IgG replacement, and vaccination timed to immune recovery (42274870, 34923107, 31753925). Rituximab should be layered into **that** programme, not prescribed with a separate ad-hoc prophylaxis plan.
- **Hypogammaglobulinaemia:** 39% developed low IgG after rituximab and 6.6% needed IVIG for recurrent sinopulmonary infection, worse with maintenance dosing (23276889, 30646343). CAR-T-specific thresholds and duration in 34757064; IgG-replacement practice is extrapolated from primary immunodeficiency, not CAR-T-derived (31416717). IVIG timing must be kept away from rituximab dosing (`REPORT.md` §4).
- **Late-onset neutropenia:** grade 3–4 neutropenia weeks to months later — incidence 3–27%, median onset 77 days (range 42–153) with infection in most (20827108); median

102 days in rheumatic disease, coinciding with the depletion window, 7/11 hospitalised with infection (21560117). Review 36706910. **[INFERENCE]** This overlaps exactly with the post-CAR-T cytopenia window, so a late neutropenia will be attributable to either agent; keeping counts is the only way to tell.

- **PML:** 57 rituximab-associated cases, median 5.5 months after the last dose, 90% fatal (19264918). New neurological signs in a CNS CAR-T patient have many likelier causes (§7), but PML belongs on the list.
- **Vaccines:** anti-CD20 blunts responses to tetanus, pneumococcal, neoantigen (KLH) and mRNA vaccines (32727835, 34514436); revaccinate after CAR-T or B-cell-depleting therapy per ASCO (38498792); CAR-T recipients lose seroprotection for specific pathogens (33914708). Vaccinate **before** depletion where the schedule allows.

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7. CAR-T-specific cautions

- **Absolute contraindication: a CD20 or RQR8 safety/selection switch in the CAR construct.** Rituximab is the agent used to delete those cells. Confirm the construct before writing the order (`REPORT.md` §5, `I_confusables_do_not_confuse`).
- Anti-CAR ADA prophylaxis with rituximab has **no** outcome data in solid-tumour CAR-T; the one in-vivo attempt (single 7 mg/kg dose 7 days pre-infusion in macaques) failed to prevent anti-CAR IgG (`REPORT.md` §2).
- Rituximab does not reverse an established titre — that is plasma-cell-mediated (`REPORT.md` §4).
- B-cell depletion could in principle blunt the CAR-T response itself (type-I-IFN-dependent effects on CD8 priming, `REPORT.md` §5); unresolved in humans.
- **New neurology after an ICV round is usually not rituximab.** TIAN (tumour inflammation-associated neurotoxicity) occurred in 16/21 children and 32% of infusions after intraventricular B7-H3 CAR-T, mostly grade 1–2, median resolution <24 h, one requiring CSF diversion (41798119); ICANS-type toxicity managed with dexamethasone and anakinra was seen with the intrathecal CART-EGFR-IL13Rα2 product used in NCT06973096 (38480922). Grade CRS/ICANS by ASTCT consensus (30592986).
- Locoregional delivery had ~61% fewer grade ≥3 adverse events than IV delivery in high-grade glioma (41895715); ICV-specific procedural AEs are mainly headache, nausea, vomiting (40282439).
- **[INFERENCE]** Rituximab does not meaningfully enter the CSF, and the CNS behaves as a B-cell sanctuary even during full peripheral depletion (42211720). A systemic dose given for systemic anti-CAR antibody prophylaxis should not be expected to deplete intrathecal B cells.

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8. Fasting, hydration, and other bedside practicalities

- **Rituximab itself does not require fasting.** The label specifies no NPO status, no food-effect restriction, and no dietary requirement — it is an IV monoclonal antibody. Patients can eat and drink before and during the infusion **[LABEL]**. Nausea/vomiting risk around the dose comes from the co-administered cyclophosphamide/fludarabine, not from rituximab.

- **Fasting belongs to the procedure, not the drug.** If a round involves sedation or general anaesthesia — Ommaya/reservoir placement, some ICV infusions, imaging under anaesthesia — the anaesthesia service's fasting rule applies on that day. Published guidance for orientation only, **the institutional anaesthesia policy governs**: ASA 2017 (2 h clear liquids, 4 h breast milk, 6 h light meal, 8 h fried/fatty meal or solids; 28045707) with the 2023 modular update on carbohydrate-containing clear liquids, chewing gum, and paediatric clear-liquid duration (36629465); ESAIC 2022 paediatric (clear fluids to **1 h**, breast milk to 3 h, gastric ultrasound as an adjunct; 34857683); ESA 2011 (21712716) **[LIT]**. Do not translate these into a fasting order for the rituximab infusion itself **[INFERENCE]**.
- **Hydration is the opposite requirement:** the label calls for aggressive IV hydration plus anti-hyperuricaemic therapy in patients at high TLS risk — which argues against casual NPO orders on rituximab day **[LABEL] [INFERENCE]**.
- **Sedating premedication:** diphenhydramine/hydroxyzine cause drowsiness; a second-generation H1 avoids it (41989666). Relevant if the same day includes a neurological examination or a procedure.
- **Time budget:** a first 375 mg/m² infusion escalating from 50 mg/h takes roughly 4–6 h plus premedication and observation; the 90-minute schedule (cycle 2 onward, steroid-containing regimens, no prior grade 3–4 IRR) is the label-sanctioned way to shorten it **[LABEL] [INFERENCE]**. Schedule accordingly on a day that also includes lymphodepletion or a procedure.
- **Chair/observation:** most reactions occur 30–120 min into the first infusion (label; 42530421), so the first dose of a course is the one requiring full monitoring capability, not the repeats.

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9. Round checklist

Before the round 1. Confirm the CAR construct has **no CD20/RQR8 switch**. 2. HBsAg, anti-HBc, anti-HBs; start antivirals and involve hepatology if either is positive (continue ≥12 months after the last dose). 3. CBC + differential + platelets, renal/liver panel, electrolytes, phosphate, uric acid, LDH; baseline quantitative IgG and B-cell flow. 4. Vaccinations up to date; no live vaccines. 5. TLS risk assessed; hydration and urate-lowering therapy ordered if high risk. 6. Baseline anti-CAR ADA sample per protocol. 7. Confirm rituximab day relative to Cy/Flu day −4 and CAR-T day 0 (day −6/−5 or day −5 per protocol), and that no procedure/anaesthesia conflicts with the infusion window.

Day of infusion 8. Acetaminophen + H1 antihistamine 30–60 min before; steroid **only** if the CAR-T protocol says so. 9. Dilute to 1–4 mg/mL in NaCl 0.9% or D5W; IV infusion only, no push/bolus. 10. First infusion 50 mg/h (paediatric 0.5 mg/kg/h, max 50 mg/h), escalate every 30 min to max 400 mg/h; subsequent infusions start 100 mg/h (paediatric 1 mg/kg/h). 11. No fasting for the infusion; encourage oral intake unless a procedure requires NPO. 12. For any reaction: stop or slow, treat by severity (steroid/epinephrine/bronchodilator/O_2), resume at ≤50% of the previous rate after resolution, discontinue for grade 3–4.

After 13. Serial CBC through the cytopenia window; consider late-onset neutropenia at 6 weeks–5 months. 14. Serial IgG; IVIG per protocol, timed away from rituximab. 15. PJP/antiviral prophylaxis integrated with the CAR-T program's plan, not a parallel one. 16.

Monitor HBV, and keep PML on the differential for new neurology alongside TIAN/ICANS and progression. 17. Anti-CAR ADA and CAR-T persistence at each redose.

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10. Sources

Label: openFDA `drug/label`, Rituxan (rituximab) IV, effective 2025-01-06 — sections 2.1 (important dosing information, infusion rates), 2.2–2.7 (indication doses), 2.8 (premedication and prophylaxis), 2.9 (administration and storage), boxed warning, 5.1–5.11.

Registry: ClinicalTrials.gov NCT04196413 (Arm D) and NCT06973096 (Cohort B), quoted in `notes/registry_and_unpublished_evidence.md`.

Literature: `index.tsv` categories `J_admin_premedication_infusion_reactions`, `K_admin_screening_prophylaxis`, `L_admin_dose_route_pk`, `M_admin_cart_context`, `N_admin_fasting_procedure`.

Search strategy, screening, and inclusion rules

Database: PubMed/Europe PMC via NCBI E-utilities (esearch/esummary/efetch), searched 2026-08-06. **3,107 unique records** were retrieved across 49 queries in 5 rounds and screened on title/abstract; 185 were curated into `../index.tsv`; 101 open-access full texts were downloaded into `../fulltext/`.

Inclusion rules

Included as **direct evidence** only if the record reports humoral anti-CAR / anti-idiotype / anti-drug antibodies (or an intervention against them) in engineered T cells used **outside** B-cell malignancy — solid tumours preferentially, plus the SIV/HIV CAR-T primate work because it is the only setting where anti-CD20 was actually given before a CAR product.

Included as **indirect/contextual**, always labelled: AAV and other gene-therapy prophylaxis; enzyme replacement and haemophilia immune tolerance induction; rituximab pharmacology; historical HAMA suppression in solid tumours; ADA assays and regulatory guidance.

Retained with an explicit `MECHANISM ONLY` flag: a small number of lymphoma/leukaemia/myeloma CAR-T papers that supply transferable biology unavailable elsewhere (first human demonstration of anti-transgene rejection; plasma-cell survival during B-cell aplasia; ADA assay validation; fully human binder redesign motivated by ADA-blocked retreatment).

Explicit exclusions

- CAR-T for lymphoma, leukaemia, myeloma, or other B-cell malignancies as a source of conclusions.
- Rituximab as anti-tumour therapy, or to sensitise/upregulate a target on tumour cells.
- Rituximab as a CAR **safety/suicide switch** (CD20, RQR8, QBEnd10) — collected separately in `I_confusables_do_not_confuse` because it is the dominant false-positive in every query combining "rituximab" with "chimeric antigen receptor", and because it is a contraindication rather than a therapy.
- Autoimmune-disease CD19 CAR-T, anti-CD20 vaccine-response studies, and generic HAMA literature, except where used purely for B-cell-depletion timing/depth principles.

Notable zero-hit queries (evidence of absence, within PubMed indexing)

- `t7_rituximab_prior_cellular_therapy` — rituximab given prior to adoptive cell therapy for immunogenicity/anti-CAR/anti-drug-antibody purposes: **0 records**.
- `t10_second_infusion_solid_tumor_outcome` — loss of expansion after a second/subsequent infusion in solid tumours, as an indexed phrase: **0 records**.

These are the two searches that would have surfaced a rituximab-plus-solid-tumour-CAR-T prophylaxis protocol if one were indexed. Absence in PubMed does not exclude unindexed trial protocols, conference abstracts, or ClinicalTrials.gov records, which were not exhaustively searched.

Screening notes

Automated keyword scoring was used for triage only; it over-ranked records where "CAR", "antibody" and "rituximab" co-occur incidentally (autoimmunity, vasculitis, vaccine, and lymphoma literature), so every curated record was confirmed by reading its title and abstract, and dose/schedule claims were taken from full text where available (`dosing_schedules_extracted.md`).

Queries

Round 1

- `q1_car_immunogenicity` `("chimeric antigen receptor" OR "CAR-T" OR "CAR T") AND (immunogenicity OR "anti-drug antibod*" OR "anti-CAR antibod*" OR "anti-idiotype" OR "anti-idiotypic" OR "neutralizing antibod*" OR "humoral immune response" OR HAMA)`
- `q2_car_rituximab_prevent` `("chimeric antigen receptor" OR "CAR-T") AND rituximab AND (immunogenicity OR "anti-drug antibod*" OR "anti-CAR" OR "B cell depletion" OR prophyla* OR tolerance)`
- `q3_car_redosing_solid` `("chimeric antigen receptor") AND ("solid tumor" OR "solid tumour") AND (redosing OR "repeat infusion" OR "repeated infusions" OR reinfusion OR "multiple doses")`
- `q4_immune_tolerance_induction_rituximab` `rituximab AND ("immune tolerance induction" OR "antibody formation" OR "inhibitor development" OR "anti-drug antibod*") AND (methotrexate OR IVIG OR "intravenous immunoglobulin" OR sirolimus OR bortezomib OR cyclophosphamide)`
- `q5_aav_gene_therapy_rituximab` `("gene therapy" OR "AAV" OR "adeno-associated virus") AND rituximab AND ("neutralizing antibod*" OR immunosuppress* OR "antibody response" OR prophylaxis)`
- `q6_cd20_safety_switch` `("chimeric antigen receptor") AND (rituximab OR CD20) AND ("safety switch" OR "suicide gene" OR RQR8 OR "elimination marker")`
- `q7_scfv_humanization_car` `("chimeric antigen receptor") AND ("murine scFv" OR humanized OR humanization OR "fully human") AND (immunogenicity OR "anti-CAR" OR rejection OR "transgene product")`

Round 2

- `a_anticar_humoral` `(("chimeric antigen receptor" OR "CAR T" OR "CAR-T" OR "engineered T cell*" OR "TCR-T" OR "transgenic TCR") AND ("anti-CAR antibod*" OR "anti-CAR immune" OR "anti-idiotyp*" OR "human anti-mouse antibod*" OR HAMA OR "antibody against the CAR" OR "antibodies against the transgene" OR "transgene immunogenicity" OR "anti-transgene immune response" OR "host immune response against" OR "immune-mediated rejection of"))`
- `b_car_ada` `(("chimeric antigen receptor" OR "CAR-T" OR "CAR T cell*") AND ("anti-drug antibody" OR "anti-drug antibodies" OR "antidrug antibod*" OR immunogenicity[Title]))`

- `c_rituximab_car_any` `rituximab AND ("chimeric antigen receptor" OR "CAR-T" OR "CAR T cell*") AND ("solid tumor*" OR "solid tumour*" OR glioma OR neuroblastoma OR sarcoma OR mesothelioma OR "renal cell" OR prostate OR ovarian OR pancrea*)`
- `d_bcell_depletion_prevent_ada` `(rituximab OR "anti-CD20" OR obinutuzumab OR ocrelizumab OR "B-cell depletion" OR "B cell depletion") AND ("prevent* the formation of" OR "prevent antibody" OR "prevention of antibody" OR "prevent* anti-drug antibod*" OR "immune tolerance induction" OR "immunosuppressive prophylaxis" OR "prophylactic immunomodulation")`
- `e_repeat_dosing_solid_car` `("chimeric antigen receptor") AND ("repeat dosing" OR "repeated dosing" OR "redosing" OR "re-dosing" OR "multiple infusions" OR "repeat infusions" OR "second infusion" OR reinfusion OR "re-infusion")`
- `f_lamers_kershaw_maus` `("carboxy-anhydrase-IX" OR "CAIX" OR "folate receptor alpha" OR mesothelin OR "HER2" OR "GD2" OR "IL13Ralpha2" OR "IL13Ra2" OR "EGFRvIII") AND ("chimeric antigen receptor") AND (immunogenic* OR "antibody response" OR anaphylax* OR "anti-idiotyp*" OR "cellular immune response against")`
- `g_aav_bcell_ritux` `("adeno-associated viral" OR "adeno-associated virus vector*" OR "AAV vector*" OR "gene transfer") AND rituximab AND (sirolimus OR "neutralizing antibod*" OR "antibody formation" OR redosing OR "immune modulation")`
- `h_ert_iti_ritux` `("enzyme replacement therapy" OR "Pompe disease" OR alglucosidase OR "hemophilia" OR "factor VIII inhibitor*") AND rituximab AND ("immune tolerance induction" OR "antibody titer*" OR prophyla*)`
- `i_ritux_dosing_pk` `rituximab AND ("dosing schedule" OR "dose schedule" OR "375 mg/m2" OR "pharmacokinetic*" AND "B cell recovery") AND ("B-cell depletion" OR "B cell depletion" OR repopulation)`
- `j_car_solid_tumor_immunosuppression` `("chimeric antigen receptor") AND ("solid tumor*" OR "solid tumour*") AND ("host immunity" OR "host immune response" OR "immune rejection" OR "anti-CAR" OR "limited persistence" OR "transgene-specific")`
- `k_mska_hama_compartmental` `(HAMA OR "human anti-mouse antibody") AND (rituximab OR immunosuppress*) AND ("radioimmunotherapy" OR "monoclonal antibody therapy" OR omburtamab OR 8H9 OR 3F8)`
- `l_car_autoimmune_ritux` `("chimeric antigen receptor") AND ("anti-CD20" OR rituximab) AND ("pretreatment" OR "pre-treatment" OR conditioning OR lymphodepletion) AND (antibod* )`

Round 3

- `s1_anticar_specific` `("anti-CAR antibod*"[tw] OR "anti-CAR immune response*"[tw] OR "anti-idiotype antibod*"[tw] AND "chimeric antigen receptor"[tw])`
- `s2_immunogenicity_cart_review` `("chimeric antigen receptor"[tw] OR "CAR T"[tw]) AND immunogenic*[ti]`
- `s3_transgene_rejection` `("chimeric antigen receptor"[tw] OR "gene-modified T cell*"[tw] OR "engineered T cell*"[tw]) AND ("transgene product"[tw] OR "transgene-specific"[tw] OR "anti-transgene"[tw] OR "immune rejection"[tw] OR "immunological rejection"[tw] OR "host-mediated rejection"[tw] OR "cellular immune response against the"[tw])`
- `s4_scfv_ada_assay` `("chimeric antigen receptor"[tw] OR "CAR T"[tw] OR "TCR-T"[tw]) AND ("anti-drug antibod*"[tw] OR "antidrug antibod*"[tw] OR "ADA assay"[tw] OR "immunogenicity assessment"[tw] OR "immunogenicity assay*"[tw])`

- `s5_solid_car_trials_repeat` `("chimeric antigen receptor"[tw]) AND ("clinical trial"[pt] OR "phase 1"[tw] OR "phase I"[ti]) AND ("solid tumor*"[tw] OR sarcoma[tw] OR glioma[tw] OR glioblastoma[tw] OR neuroblastoma[tw] OR mesothelioma[tw] OR "renal cell carcinoma"[tw] OR "prostate cancer"[tw] OR "ovarian cancer"[tw] OR "pancreatic cancer"[tw] OR "colorectal"[tw] OR "pontine glioma"[tw] OR "midline glioma"[tw]) AND ("repeat"[tw] OR "repeated"[tw] OR "multiple infusions"[tw] OR "second infusion"[tw] OR redos*[tw] OR reinfus*[tw] OR "re-infusion"[tw] OR "intracranial"[tw] OR "intraperitoneal"[tw] OR "intrapleural"[tw] OR "hepatic artery"[tw])`
- `s6_ritux_prophylaxis_gt` `rituximab[tw] AND ("gene therapy"[tw] OR "gene transfer"[tw] OR "adeno-associated"[tw] OR "AAV vector"[tw] OR "enzyme replacement"[tw] OR "factor VIII"[tw] OR "factor IX"[tw] OR "asparaginase"[tw] OR "immune tolerance induction"[tw]) AND (prophyla*[tw] OR prevent*[tw] OR "antibody formation"[tw] OR "antibody titer*"[tw] OR "anti-drug antibod*"[tw] OR "neutralizing antibod*"[tw] OR "inhibitor development"[tw])`
- `s7_ritux_bcell_kinetics` `rituximab[tw] AND ("B-cell depletion"[tw] OR "B cell depletion"[tw] OR repopulation[tw] OR reconstitution[tw]) AND (pharmacokinetic*[tw] OR "dose"[ti] OR dosing[tw] OR "375 mg"[tw] OR "1000 mg"[tw] OR schedule*[tw]) AND ("plasma cell*"[tw] OR "memory B cell*"[tw] OR "antibody response"[tw] OR "serum immunoglobulin"[tw] OR "vaccine response"[tw] OR duration[tw])`
- `s8_ritux_kill_switch` `(rituximab[tw] OR "CD20"[tw]) AND ("safety switch"[tw] OR "suicide switch"[tw] OR "elimination marker"[tw] OR RQR8[tw] OR "QBEnd10"[tw]) AND ("chimeric antigen receptor"[tw] OR "T cell*"[tw])`
- `s9_hama_solid_mab` `("human anti-mouse antibod*"[tw] OR HAMA[tw] OR "human anti-chimeric antibod*"[tw]) AND (neuroblastoma[tw] OR glioma[tw] OR "solid tumor*"[tw] OR "radioimmunotherapy"[tw] OR "3F8"[tw] OR "8H9"[tw] OR omburtamab[tw] OR dinutuximab[tw] OR "ch14.18"[tw])`
- `s10_bcell_depletion_before_immunogen` `(rituximab[tw] OR "anti-CD20"[tw]) AND ("prior to"[tw] OR pretreat*[tw] OR "pre-treatment"[tw] OR preconditioning[tw] OR "before administration"[tw]) AND ("prevent* antibod*"[tw] OR "blunt* the antibody"[tw] OR "abrogat* antibod*"[tw] OR "suppress* antibody"[tw] OR "humoral response"[tw] OR "primary antibody response"[tw])`
- `s11_car_solid_persistence_host` `("chimeric antigen receptor"[tw]) AND ("solid tumor*"[tw] OR "solid tumour*"[tw]) AND ("limited persistence"[tw] OR "lack of persistence"[tw] OR "poor persistence"[tw] OR "host immune response*"[tw] OR "immunogenicity"[tw])`
- `s12_ivig_plasma_cell_ada` `("anti-drug antibod*"[tw] OR "neutralizing antibod*"[tw] OR "inhibitor*"[tw]) AND (bortezomib[tw] OR daratumumab[tw] OR "anti-CD38"[tw] OR "plasma cell depletion"[tw] OR "IdeS"[tw] OR imlifidase[tw] OR "anti-CD20"[tw] OR rituximab[tw]) AND ("gene therapy"[tw] OR "AAV"[tw] OR "enzyme replacement"[tw] OR "cell therapy"[tw] OR "redos*"[tw] OR "re-administration"[tw] OR readministration[tw])`

Round 4

- `t1_lamers` `Lamers CH[au] AND ("chimeric"[tw] OR CAIX[tw] OR "carboxy-anhydrase"[tw] OR "renal cell"[tw])`
- `t2_kershaw_frα` `("folate receptor"[tw]) AND ("chimeric receptor"[tw] OR "chimeric antigen receptor"[tw]) AND (ovarian[tw] OR "phase I"[tw] OR "clinical trial"[pt])`
- `t3_anticd20_prevent_anticar_preclin` `("anti-CD20"[tw] OR rituximab[tw] OR "B cell depletion"[tw] OR "B-cell depletion"[tw] OR "BAFF"[tw] OR "anti-CD38"[tw] OR`

daratumumab[tw]) AND ("adoptive transfer"[tw] OR "adoptive cell"[tw] OR "CAR T"[tw] OR "chimeric antigen receptor"[tw] OR "gene-modified T cell*"[tw]) AND ("anti-CAR"[tw] OR "anti-transgene"[tw] OR "antibody response"[tw] OR "humoral response"[tw] OR "repeat dosing"[tw] OR readministration[tw] OR "re-administration"[tw] OR persistence[tw])

- `t4_hama_prevent_immunosuppression` `(HAMA[tw] OR "human anti-mouse antibod*"[tw] OR "human antimouse antibod*"[tw]) AND (prevent*[tw] OR abrogat*[tw] OR suppress*[tw] OR rituximab[tw] OR cyclosporin*[tw] OR "immunosuppress*"[tw] OR methotrexate[tw] OR "deoxyspergualin"[tw]) AND ("repeat*"[tw] OR "retreatment"[tw] OR "multiple cycles"[tw] OR "subsequent"[tw] OR "murine antibod*"[tw])`
- `t5_aav_readmin_immunomod` `("adeno-associated"[tw] OR "AAV"[ti]) AND (readministration[tw] OR "re-administration"[tw] OR redosing[tw] OR "repeat administration"[tw] OR "second dose"[tw]) AND (rituximab[tw] OR "B cell depletion"[tw] OR imlifidase[tw] OR IdeS[tw] OR plasmapheresis[tw] OR bortezomib[tw] OR "immune modulation"[tw] OR sirolimus[tw] OR rapamycin[tw])`
- `t6_solid_car_trials_immunogenicity_named` `("chimeric antigen receptor"[tw]) AND ("HER2"[tw] OR "GD2"[tw] OR "B7-H3"[tw] OR "IL13Ralpha2"[tw] OR "IL13Ra2"[tw] OR "EGFRvIII"[tw] OR mesothelin[tw] OR "CEA"[tw] OR "PSCA"[tw] OR "PSMA"[tw] OR CLDN6[tw] OR "claudin 18.2"[tw] OR GPC3[tw] OR "CAIX"[tw] OR "MUC1"[tw] OR "CD70"[tw]) AND ("phase 1"[tw] OR "phase I"[ti] OR "first-in-human"[tw] OR "clinical trial"[pt]) AND ("solid"[tw] OR glioma[tw] OR sarcoma[tw] OR neuroblastoma[tw] OR carcinoma[tw] OR glioblastoma[tw] OR mesothelioma[tw])`
- `t7_rituximab_prior_cellular_therapy` `rituximab[tw] AND ("adoptive cell therapy"[tw] OR "adoptive immunotherapy"[tw] OR "cell therapy"[tw] OR "CAR T"[tw]) AND ("prior to"[tw] OR before[tw] OR prophylactic[tw] OR preemptive[tw]) AND ("antibody formation"[tw] OR "anti-drug antibod*"[tw] OR "immunogenicity"[tw] OR "anti-CAR"[tw] OR "neutralizing"[tw])`
- `t8_ritux_pk_dose_response` `rituximab[tw] AND ("low dose"[tw] OR "single dose"[tw] OR "100 mg"[tw] OR "375 mg/m2"[tw] OR "1000 mg"[tw] OR "dose-finding"[tw]) AND ("B cell depletion"[tw] OR "B-cell depletion"[tw]) AND ("kinetics"[tw] OR duration[tw] OR "time to"[tw] OR onset[tw] OR "lymph node"[tw] OR "germinal cent*"[tw] OR "tissue"[tw])`
- `t9_regulatory_guidance_immunogenicity` `("immunogenicity"[ti] OR "anti-drug antibod*"[ti]) AND (guidance[tw] OR "regulatory"[tw] OR "FDA"[tw] OR "EMA"[tw] OR "risk assessment"[tw]) AND ("cell therapy"[tw] OR "gene therapy"[tw] OR "CAR-T"[tw] OR "CAR T"[tw])`
- `t10_second_infusion_solid_tumor_outcome` `("chimeric antigen receptor"[tw] OR "CAR-T"[tw]) AND ("solid"[tw] OR glioma[tw] OR sarcoma[tw] OR neuroblastoma[tw] OR mesothelioma[tw] OR carcinoma[tw]) AND ("loss of"[tw] OR "decreased"[tw] OR "diminished"[tw] OR "absent"[tw]) AND ("expansion after the second"[tw] OR "subsequent infusions"[tw] OR "second infusion"[tw] OR "later infusions"[tw] OR "repeat infusions"[tw])`

Round 5

- `u1` `Kershaw MH[au] AND ("folate receptor"[tw] OR "phase I"[tw] OR ovarian[tw])`
- `u2` `Ahmed N[au] AND "HER2"[tw] AND (sarcoma[tw] OR glioblastoma[tw]) AND ("chimeric antigen receptor"[tw] OR "virus-specific"[tw])`
- `u3` `Beatty GL[au] AND mesothelin[tw] AND (mRNA[tw] OR "chimeric antigen receptor"[tw])`
- `u4` `("PSCA"[tw] OR "prostate stem cell antigen"[tw]) AND "chimeric antigen receptor"[tw] AND ("phase 1"[tw] OR trial[tw])`

- `u5 Thistlethwaite FC[au] OR ("CEA"[tw] AND "chimeric antigen receptor"[tw] AND ("phase I"[tw] OR "first-in-man"[tw]) AND (colorectal[tw] OR "carcinoembryonic"[tw]))`
- `u6 ("anti-CD20"[tw] OR rituximab[tw]) AND ("KLH"[tw] OR "keyhole limpet"[tw] OR "neoantigen challenge"[tw] OR "primary immunization"[tw] OR "de novo antibody"[tw] OR "T-dependent antigen"[tw])`
- `u7 ("germinal center"[tw] OR "lymph node"[tw] OR "tissue-resident"[tw] OR spleen[tw]) AND (rituximab[tw] OR "anti-CD20"[tw]) AND ("incomplete depletion"[tw] OR "residual B cells"[tw] OR "depletion of B cells"[tw] OR "resistance to depletion"[tw])`
- `u8 ("chimeric antigen receptor"[tw] OR "CAR T"[tw]) AND ("cyclophosphamide"[tw] AND "fludarabine"[tw]) AND ("lymphodepletion"[tw]) AND ("B cell"[tw] OR "humoral"[tw] OR "antibody"[tw] OR "immunogenicity"[tw])`
- `u9 ("mycophenolate"[tw] OR "tacrolimus"[tw] OR "sirolimus"[tw] OR "rapamycin"[tw] OR "belatacept"[tw] OR "abatacept"[tw] OR "CTLA4-Ig"[tw]) AND ("anti-drug antibod*"[tw] OR "antibody formation"[tw] OR "immune tolerance"[tw]) AND ("gene therapy"[tw] OR "cell therapy"[tw] OR "transgene"[tw] OR "biologic*"[tw])`
- `u10 ("humanized"[tw] OR "fully human"[tw]) AND "chimeric antigen receptor"[tw] AND (retreatment[tw] OR "second infusion"[tw] OR "anti-CAR"[tw] OR "anti-murine"[tw] OR immunogenic*[tw]) AND ("solid"[tw] OR glioma[tw] OR sarcoma[tw] OR neuroblastoma[tw] OR myeloma[tw])`

Administration-level supplementary search (categories J–N)

Second pass, run to answer the follow-up question "how is the drug actually given" rather than "does it work". Source: Europe PMC REST (`/search`, `resultType=core`), no API key; script `harvest_administration.py`, 1,365 unique records across seven query groups, then hand-screened down to the 59 records now in `index.tsv` categories `J–N`. Metadata and abstracts refetched by PMID with `curate_administration.py`; rows appended and PMC open-access XML fetched by `index_administration.py`.

Query groups (full strings in `harvest_administration.py`):

Group	Intent
`premedication_infusion_reactions`	premedication components, rapid/shortened infusion schedules, IRR incidence and management, desensitisation, anti-rituximab antibodies
`pediatric_administration`	paediatric infusion rates, PK/BSA dosing, hypogammaglobulinaemia in children
`screening_and_prophylaxis`	HBV, PJP, IgG replacement, vaccination, late-onset neutropenia, TLS, PML
`cns_and_route`	CSF penetration, intrathecal/intraventricular routes, subcutaneous and biosimilar formulations
`dose_depth_duration`	dose–depletion–duration, high-dose (500–750 mg/m²) exposure and safety
`cart_context`	steroids and CAR-T function, Cy/Flu lymphodepletion dosing, ICV procedure, infection prophylaxis guidelines
`fasting_npo_sedation`	preoperative/procedural fasting guidelines, whether any infusion requires fasting

Non-literature sources used for the same note: openFDA `drug/label` (Rituxan IV, label effective 2025-01-06) for all label-derived statements, and the two ClinicalTrials.gov records already quoted in `notes/registry_and_unpublished_evidence.md`. The searches returned **no** publication specifying premedication, infusion rate, or fasting requirements for rituximab given for anti-CAR prophylaxis in solid-tumour CAR-T; every administration statement in `notes/administration_protocol.md` is therefore label-derived, indication-transferred, or explicitly labelled INFERENCE.

Appendix: full reference list

All 252 curated records, grouped by category and ordered by year (descending). Records without a PMID are trial-registry entries, preprints, or presentations.

A — Anti-CAR antibody evidence in solid tumours (18)

- Tebas P et al. Clinical trial results provide the rationale to protect dual HIV-specific T cells with a signaling-defective HIV fusion inhibitor. Mol Ther 2026. PMID 41520175.
- Chen Y et al. Anti-CAR Immunity Drives Acquired Therapeutic Resistance to GD2-CAR T Cell Therapy in Diffuse Midline Glioma. medRxiv (preprint) 2026. PMID 42465905.
- Kimble EL et al. Novel antibodies for identification, selection, and manipulation of T cells expressing Whitlow linker-containing CARs. J Immunother Cancer 2025. PMID 41253492.
- Qi C et al. Claudin-18 isoform 2-specific CAR T-cell therapy (satri-cel) versus treatment of physician's choice for previously treated advanced gastric or gastro-oesophageal junction cancer (CT041-ST-01): a randomised, open-label, phase 2 trial. Lancet 2025. PMID 40460847.
- Pampusch MS et al. Assessment of anti-CD20 antibody pre-treatment for augmentation of CAR-T cell therapy in SIV-infected rhesus macaques. Front Immunol 2023. PMID 36825014.
- Davey BC et al. Development of an anti-CAR antibody response in SIV-infected rhesus macaques treated with CD4-MBL CAR/CXCR5 T cells. Front Immunol 2022. PMID 36582226.
- Ko AH et al. Dual Targeting of Mesothelin and CD19 with Chimeric Antigen Receptor-Modified T Cells in Patients with Metastatic Pancreatic Cancer. Mol Ther 2020. PMID 32730744.
- Haas AR et al. Phase I Study of Lentiviral-Transduced Chimeric Antigen Receptor-Modified T Cells Recognizing Mesothelin in Advanced Solid Cancers. Mol Ther 2019. PMID 31420241.
- Hege KM et al. Safety, tumor trafficking and immunogenicity of chimeric antigen receptor (CAR)-T cells specific for TAG-72 in colorectal cancer. J Immunother Cancer 2017. PMID 28344808.
- Lamers CH et al. Treatment of metastatic renal cell carcinoma (mRCC) with CAIX CAR-engineered T-cells-a completed study overview. Biochem Soc Trans 2016. PMID 27284065.
- Junghans RP et al. Phase I Trial of Anti-PSMA Designer CAR-T Cells in Prostate Cancer: Possible Role for Interacting Interleukin 2-T Cell Pharmacodynamics as a Determinant of Clinical Response. Prostate 2016. PMID 27324746.
- Beatty GL et al. Mesothelin-specific chimeric antigen receptor mRNA-engineered T cells induce anti-tumor activity in solid malignancies. Cancer Immunol Res 2014. PMID 24579088.
- Lamers CH et al. Treatment of metastatic renal cell carcinoma with CAIX CAR-engineered T cells: clinical evaluation and management of on-target toxicity. Mol Ther 2013. PMID 23423337.

- Maus MV et al. T cells expressing chimeric antigen receptors can cause anaphylaxis in humans. Cancer Immunol Res 2013. PMID 24777247.
- Lamers CH et al. Immune responses to transgene and retroviral vector in patients treated with ex vivo-engineered T cells. Blood 2011. PMID 20889925.
- Lamers CH et al. Gene-modified T cells for adoptive immunotherapy of renal cell cancer maintain transgene-specific immune functions in vivo. Cancer Immunol Immunother 2007. PMID 17479266.
- Lamers CH et al. Treatment of metastatic renal cell carcinoma with autologous T-lymphocytes genetically retargeted against carbonic anhydrase IX: first clinical experience. J Clin Oncol 2006. PMID 16648493.
- Kershaw MH et al. A phase I study on adoptive immunotherapy using gene-modified T cells for ovarian cancer. Clin Cancer Res 2006. PMID 17062687.

B — Mechanism, ADA assays, regulatory guidance (28)

- Hofman K et al. Immunogenicity of Gene and Cell Therapies. BioDrugs 2026. PMID 41518483.
- Day G et al. Development of a flow cytometry method to measure antidrug antibodies against CAR T cells. Immunohorizons 2026. PMID 42033387.
- Liu X et al. Epitope mapping of humoral immunogenicity of orvacabtagene autoleucel shows an IgM response with minimal impact on CAR T cellular kinetics. Mol Ther Adv 2026. PMID 42375474.
- Fernández N et al. A roadmap for systematic humanization of a chimeric antigen receptor: preclinical validation of a humanized CD22 scFv as a model. Exp Hematol 2026. PMID 41881100.
- Alfar HR et al. Clinical evidence of immunogenicity of CAR-T cell therapies and its implication in the clinical development of CAR-T drug products. Front Immunol 2025. PMID 40061940.
- Masilamani M et al. Cellular Immunogenicity Assessments in CAR-T Cell Therapies: Current Insights and Future Directions. AAPS J 2025. PMID 40908464.
- Gokemeijer J et al. An IQ Consortium Perspective on Best Practices for Bioanalytical and Immunogenicity Assessment Aspects of CAR-T and TCR-T Cellular Therapies Development. Clin Pharmacol Ther 2024. PMID 37983584.
- Mattei AE et al. In silico methods for immunogenicity risk assessment and human homology screening for therapeutic antibodies. MAbs 2024. PMID 38536724.
- Sauna ZE et al. Understanding preclinical and clinical immunogenicity risks in novel biotherapeutics development. Front Immunol 2023. PMID 37251396.
- Khan AN et al. Immunogenicity of CAR-T Cell Therapeutics: Evidence, Mechanism and Mitigation. Front Immunol 2022. PMID 35677038.
- Wagner DL et al. Immunogenicity of CAR T cells in cancer therapy. Nat Rev Clin Oncol 2021. PMID 33633361.
- Zinsli LV et al. Deimmunization of protein therapeutics - Recent advances in experimental and computational epitope prediction and deletion. Comput Struct Biotechnol J 2021. PMID 33425259.
- Thudium Mueller K et al. Tisagenlecleucel immunogenicity in relapsed/refractory acute lymphoblastic leukemia and diffuse large B-cell lymphoma. Blood Adv 2021. PMID 34432863.
- Gazeau N et al. Effective anti-BCMA retreatment in multiple myeloma. Blood Adv 2021. PMID 34351389.
- Wang D et al. A phase 1 study of a novel fully human BCMA-targeting CAR (CT103A) in patients with relapsed/refractory multiple myeloma. Blood 2021. PMID 33512480.
- Potthoff B et al. A cell-based immunogenicity assay to detect antibodies against chimeric antigen receptor expressed by tisagenlecleucel. J Immunol Methods 2020. PMID 31678267.
- Awasthi R et al. Tisagenlecleucel cellular kinetics, dose, and immunogenicity in relation to clinical factors in relapsed/refractory DLBCL. Blood Adv 2020. PMID 32045475.
- Gorovits B et al. Immunogenicity of Chimeric Antigen Receptor T-Cell Therapeutics. BioDrugs 2019. PMID 31069709.
- Davda J et al. Immunogenicity of immunomodulatory, antibody-based, oncology therapeutics. J Immunother Cancer 2019. PMID 30992085.
- Xu J et al. Exploratory trial of a biepitopic CAR T-targeting B cell maturation antigen in relapsed/refractory multiple myeloma. Proc Natl Acad Sci U S A 2019. PMID 30988175.
- Salazar-Fontana LI et al. Approaches to Mitigate the Unwanted Immunogenicity of Therapeutic Proteins during Drug Development. AAPS J 2017. PMID 28083796.
- Turtle CJ et al. CD19 CAR-T cells of defined CD4+:CD8+ composition in adult B cell ALL patients. J Clin Invest 2016. PMID 27111235.
- Song DG et al. A fully human chimeric antigen receptor with potent activity against cancer cells but reduced risk for off-tumor toxicity. Oncotarget 2015. PMID 26101914.

- Lanitis E et al. Redirected antitumor activity of primary human lymphocytes transduced with a fully human anti-mesothelin chimeric receptor. Mol Ther 2012. PMID 22127019.
- Jensen MC et al. Antitransgene rejection responses contribute to attenuated persistence of adoptively transferred CD20/CD19-specific chimeric antigen receptor redirected T cells in humans. Biol Blood Marrow Transplant 2010. PMID 20304086.
- Piasecki JC et al. Induction of transgene-specific cytotoxic T lymphocyte responses after transplantation of gene-modified CD34+ cells despite nonablative immunosuppressive conditioning. Hum Gene Ther 2008. PMID 18092920.
- Berger C et al. Analysis of transgene-specific immune responses that limit the in vivo persistence of adoptively transferred HSV-TK-modified donor T cells after allogeneic hematopoietic cell transplantation. Blood 2006. PMID 16282341.
- Berger C et al. Nonmyeloablative immunosuppressive regimen prolongs In vivo persistence of gene-modified autologous T cells in a nonhuman primate model. J Virol 2001. PMID 11134293.

C — Solid-tumour CAR-T trials that redose (40)

- Shan Y et al. T-cell receptor-like chimeric antigen receptor T cells targeting mesothelin: A first-in-human dose-escalation trial for platinum-resistant advanced ovarian cancer. Cancer 2026. PMID 41575866.
- Ronsley R et al. Intracerebroventricular B7-H3-targeting CAR T cells for non-pontine DMG and recurrent/refractory pediatric CNS tumors: a phase 1 trial. Neuro Oncol 2026. PMID 42503899.
- Gao Y et al. Hypoxia-responsive CEA-targeted CAR T cells in CEA-positive solid tumors through intraperitoneal or intravenous infusion: a phase 1 trial. Nat Cancer 2026. PMID 41760800.
- Liu C et al. Allogeneic B7-H3-Targeted CAR Vδ1T-cell Therapy in Advanced Solid Tumors: A Phase I Study. Clin Cancer Res 2026. PMID 41779003.
- Srour SA et al. Allogeneic CD70-Targeted Chimeric Antigen Receptor T-Cell Therapy for Advanced Renal Cell Carcinoma: Results From the Phase I TRAVERSE Trial. J Clin Oncol 2026. PMID 42447427.
- Mackall CL. GD2-CAR T Cells for Diffuse Midline Gliomas (CLIN2-12595), CIRM Board Meeting presentation, January 2026. CIRM Board Meeting (presentation) 2026. https://www.cirm.ca.gov/wp-content/uploads/2026/01/7b.-20290129-Closer-to-Cures-Presentation.pdf.
- Stanford University (sponsor). GD2 CAR T Cells in Diffuse Intrinsic Pontine Gliomas (DIPG) & Spinal Diffuse Midline Glioma (DMG) - NCT04196413, Arm D. ClinicalTrials.gov 2026. https://clinicaltrials.gov/study/NCT04196413.
- Freeburg NF et al. The critical role of the endogenous immune compartment after CAR T cell therapy in recurrent GBM. Cell 2026. PMID 42296961.
- Monje M et al. Intravenous and intracranial GD2-CAR T cells for H3K27M+ diffuse midline gliomas. Nature 2025. PMID 39537919.
- Vitanza NA et al. Intracerebroventricular B7-H3-targeting CAR T cells for diffuse intrinsic pontine glioma: a phase 1 trial. Nat Med 2025. PMID 39775044.
- Barish ME et al. Chlorotoxin-directed CAR T cell therapy for recurrent glioblastoma: Interim clinical experience demonstrating feasibility and safety. Cell Rep Med 2025. PMID 40818458.
- Li B et al. Intraperitoneal infusion of NKG2D CAR-NK cells induces endogenous CD8+ T cell activation in patients with advanced colorectal cancer. Mol Ther 2025. PMID 40437757.
- Begley SL et al. CAR T cell therapy for glioblastoma: A review of the first decade of clinical trials. Mol Ther 2025. PMID 40057825.
- Silbert S et al. Have CARs stalled for non-B-cell malignancies? Where are we, and where are we going? Hematology Am Soc Hematol Educ Program 2025. PMID 41347986.
- University of Pennsylvania (sponsor). CART-EGFR-IL13Ra2 in Newly Diagnosed GBM Following Initial Radiotherapy - NCT06973096, Cohort B. ClinicalTrials.gov 2025. https://clinicaltrials.gov/study/NCT06973096.
- Bagley SJ et al. Intracerebroventricular bivalent CAR T cells targeting EGFR and IL-13Ralpha2 in recurrent glioblastoma: a phase 1 trial. Nat Med 2025. PMID 40451950.
- Dorff TB et al. PSCA-CAR T cell therapy in metastatic castration-resistant prostate cancer: a phase 1 trial. Nat Med 2024. PMID 38867077.
- Stein MN et al. PSCA-targeted BPX-601 CAR T cells with pharmacological activation by rimiducid in metastatic pancreatic and prostate cancer: a phase 1 dose escalation trial. Nat Commun 2024. PMID 39737899.
- Hegde M et al. Autologous HER2-specific CAR T cells after lymphodepletion for advanced sarcoma: a phase 1 trial. Nat Cancer 2024. PMID 38658775.
- Bagley SJ et al. Repeated peripheral infusions of anti-EGFRvIII CAR T cells in combination with pembrolizumab show no efficacy in glioblastoma: a phase 1 trial. Nat Cancer 2024. PMID 38216766.

- Pinto N et al. STRIvE-02: A First-in-Human Phase I Study of Systemically Administered B7-H3 Chimeric Antigen Receptor T Cells for Patients With Relapsed/Refractory Solid Tumors. J Clin Oncol 2024. PMID 39255444.
- Mackensen A et al. CLDN6-specific CAR-T cells plus amplifying RNA vaccine in relapsed or refractory solid tumors: the phase 1 BNT211-01 trial. Nat Med 2023. PMID 37872225.
- Majzner RG et al. GD2-CAR T cell therapy for H3K27M-mutated diffuse midline gliomas. Nature 2022. PMID 35130560.
- Brown CE et al. Off-the-shelf, steroid-resistant, IL13Rα2-specific CAR T cells for treatment of glioblastoma. Neuro Oncol 2022. PMID 35100373.
- Ghosn M et al. Image-guided interventional radiological delivery of chimeric antigen receptor (CAR) T cells for pleural malignancies in a phase I/II clinical trial. Lung Cancer 2022. PMID 35045358.
- Vitanza NA et al. Locoregional infusion of HER2-specific CAR T cells in children and young adults with recurrent or refractory CNS tumors: an interim analysis. Nat Med 2021. PMID 34253928.
- Adusumilli PS et al. A Phase I Trial of Regional Mesothelin-Targeted CAR T-cell Therapy in Patients with Malignant Pleural Disease, in Combination with the Anti-PD-1 Agent Pembrolizumab. Cancer Discov 2021. PMID 34266984.
- Katz SC et al. HITM-SURE: Hepatic immunotherapy for metastases phase Ib anti-CEA CAR-T study utilizing pressure enabled drug delivery. J Immunother Cancer 2020. PMID 32843493.
- Katz SC et al. HITM-SIR: phase Ib trial of intraarterial chimeric antigen receptor T-cell therapy and selective internal radiation therapy for CEA+ liver metastases. Cancer Gene Ther 2020. PMID 31155611.
- Beatty GL et al. Activity of Mesothelin-Specific Chimeric Antigen Receptor T Cells Against Pancreatic Carcinoma Metastases in a Phase 1 Trial. Gastroenterology 2018. PMID 29567081.
- Wang Y et al. CD133-directed CAR T cells for advanced metastasis malignancies: A phase I trial. Oncoimmunology 2018. PMID 29900044.
- Thistlethwaite FC et al. The clinical efficacy of first-generation carcinoembryonic antigen (CEACAM5)-specific CAR T cells is limited by poor persistence and transient pre-conditioning-dependent respiratory toxicity. Cancer Immunol Immunother 2017. PMID 28660319.
- Zhang C et al. Phase I Escalating-Dose Trial of CAR-T Therapy Targeting CEA+ Metastatic Colorectal Cancers. Mol Ther 2017. PMID 28366766.
- Ahmed N et al. HER2-Specific Chimeric Antigen Receptor-Modified Virus-Specific T Cells for Progressive Glioblastoma: A Phase 1 Dose-Escalation Trial. JAMA Oncol 2017. PMID 28426845.
- Brown CE et al. Regression of Glioblastoma after Chimeric Antigen Receptor T-Cell Therapy. N Engl J Med 2016. PMID 28029927.
- Katz SC et al. Regional CAR-T cell infusions for peritoneal carcinomatosis are superior to systemic delivery. Cancer Gene Ther 2016. PMID 27080226.
- Ahmed N et al. Human Epidermal Growth Factor Receptor 2 (HER2) -Specific Chimeric Antigen Receptor-Modified T Cells for the Immunotherapy of HER2-Positive Sarcoma. J Clin Oncol 2015. PMID 25800760.
- Brown CE et al. Bioactivity and Safety of IL13Rα2-Redirected Chimeric Antigen Receptor CD8+ T Cells in Patients with Recurrent Glioblastoma. Clin Cancer Res 2015. PMID 26059190.
- Adusumilli PS et al. Regional delivery of mesothelin-targeted CAR T cell therapy generates potent and long-lasting CD4-dependent tumor immunity. Sci Transl Med 2014. PMID 25378643.
- Saied A et al. Neutrophil:lymphocyte ratios and serum cytokine changes after hepatic artery chimeric antigen receptor-modified T-cell infusions for liver metastases. Cancer Gene Ther 2014. PMID 25277132.

D — Rituximab pharmacology and what sets a schedule (26)

- Chen C et al. The rituximab paradox in rheumatoid arthritis: Re-evaluating B-cell depletion depth and the synovial microenvironment. Autoimmun Rev 2026. PMID 42162633.
- Yuan M et al. Translating B-Cell and Plasma-Cell Targeting from Oncology to Autoimmunity: Modalities, Quantitative Bridging, and a Development Roadmap. Clin Pharmacol Ther 2026. PMID 42244451.
- Cao S et al. Efficacy, safety, and B-cell depletion capacity of 3 rituximab dosing regimens in the treatment of moderate-to-severe pemphigus vulgaris and pemphigus foliaceus: A 52-week clinical trial. J Am Acad Dermatol 2025. PMID 40374120.
- Liang H et al. Dosing optimization of rituximab for primary membranous nephropathy by population pharmacokinetic and pharmacodynamic study. Front Pharmacol 2024. PMID 38595918.
- Ciolfi C et al. Is It Time to Reconsider Rituximab Dosing Regimens for Pemphigus Vulgaris? Antibodies (Basel) 2024. PMID 38247568.

- Hamaya T et al. Humoral response to SARS-CoV-2 mRNA vaccine on in ABO blood type incompatible kidney transplant recipients treated with low-dose rituximab. Sci Rep 2023. PMID 37699969.
- Shree T et al. CD20-Targeted Therapy Ablates De Novo Antibody Response to Vaccination but Spares Preestablished Immunity. Blood Cancer Discov 2022. PMID 35015688.
- Blincoe A et al. Acquired B-cell deficiency secondary to B-cell-depleting therapies. J Immunol Methods 2022. PMID 36372267.
- van Dam LS et al. Highly Sensitive Flow Cytometric Detection of Residual B-Cells After Rituximab in Anti-Neutrophil Cytoplasmic Antibodies-Associated Vasculitis Patients. Front Immunol 2020. PMID 33384685.
- Ramwadhdoebe TH et al. Effect of rituximab treatment on T and B cell subsets in lymph node biopsies of patients with rheumatoid arthritis. Rheumatology (Oxford) 2019. PMID 30649469.
- Hogan J et al. Effect of different rituximab regimens on B cell depletion and time to relapse in children with steroid-dependent nephrotic syndrome. Pediatr Nephrol 2019. PMID 30109447.
- Redfield RR et al. Safety, pharmacokinetics, and pharmacodynamic activity of obinutuzumab, a type 2 anti-CD20 monoclonal antibody for the desensitization of candidates for renal transplant. Am J Transplant 2019. PMID 31257724.
- Cornec D et al. Pharmacokinetics of rituximab and clinical outcomes in patients with anti-neutrophil cytoplasmic antibody associated vasculitis. Rheumatology (Oxford) 2018. PMID 29340623.
- Bhoj VG et al. Persistence of long-lived plasma cells and humoral immunity in individuals responding to CD19-directed CAR T-cell therapy. Blood 2016. PMID 27166358.
- Laws LH et al. Inflammation Causes Resistance to Anti-CD20-Mediated B Cell Depletion. Am J Transplant 2016. PMID 27265023.
- Rosenberg AS et al. A role for plasma cell targeting agents in immune tolerance induction in autoimmune disease and antibody responses to therapeutic proteins. Clin Immunol 2016. PMID 26928739.
- Turner-Stokes T et al. Induction treatment of ANCA-associated vasculitis with a single dose of rituximab. Rheumatology (Oxford) 2014. PMID 24609057.
- Vaidyanathan A et al. Developmental immunotoxicology assessment of rituximab in cynomolgus monkeys. Toxicol Sci 2011. PMID 20937725.
- Bingham CO et al. Immunization responses in rheumatoid arthritis patients treated with rituximab: results from a controlled clinical trial. Arthritis Rheum 2010. PMID 20039397.
- St Clair EW. Good and bad memories following rituximab therapy. Arthritis Rheum 2010. PMID 20039422.
- Bearden CM et al. Rituximab inhibits the in vivo primary and secondary antibody response to a neoantigen, bacteriophage phiX174. Am J Transplant 2005. PMID 15636611.
- Vugmeyster Y et al. Depletion of B cells by a humanized anti-CD20 antibody PRO70769 in Macaca fascicularis. J Immunother 2005. PMID 15838377.
- Schröder C et al. Anti-CD20 treatment depletes B-cells in blood and lymphatic tissue of cynomolgus monkeys. Transpl Immunol 2003. PMID 14551029.
- van der Kolk LE et al. Rituximab treatment results in impaired secondary humoral immune responsiveness. Blood 2002. PMID 12200395.
- Gonzalez-Stawinski GV et al. Hapten-induced primary and memory humoral responses are inhibited by the infusion of anti-CD20 monoclonal antibody (IDEC-C2B8, Rituximab). Clin Immunol 2001. PMID 11161973.
- Maloney DG et al. IDEC-C2B8 (Rituximab) anti-CD20 monoclonal antibody therapy in patients with relapsed low-grade non-Hodgkin's lymphoma. Blood 1997. PMID 9310469.

E — Anti-CD20 prophylaxis in gene therapy (20)

- Vicidomini A et al. Rapamycin nanoparticles mitigate anti-AAV antibody formation in a mouse model of ornithine transcarbamylase deficiency. Mol Ther Adv 2026. PMID 42436852.
- Li L et al. Cholesterolated Rapamycin Prodrug Liposomes Induce Antigen-Specific Tolerance and Enable AAV Redosing. Mol Pharm 2026. PMID 41674289.
- Doshi BS et al. Use of CD19-targeted immune modulation to eradicate AAV-neutralizing antibodies. Mol Ther 2025. PMID 40057826.
- Nelson RW et al. Reduction of Preexisting AAV9 Antibody Titers Before Onasemnogene Abeparvovec Administration in Twins With Spinal Muscular Atrophy. Neurology 2025. PMID 40705999.
- Li X et al. Lipid-Rapamycin Nanovaccines Overcome the Antidrug Antibody Barrier in Biologic Therapies. ACS Nano 2025. PMID 39847793.
- Rana J et al. B cell focused transient immune suppression protocol for efficient AAV readministration to the liver. Mol Ther Methods Clin Dev 2024. PMID 38440160.

- Potter RA et al. Use of plasmapheresis to lower anti-AAV antibodies in nonhuman primates with pre-existing immunity to AAVrh74. Mol Ther Methods Clin Dev 2024. PMID 38327805.
- Salabarria SM et al. Thrombotic microangiopathy following systemic AAV administration is dependent on anti-capsid antibodies. J Clin Invest 2024. PMID 37988172.
- Choi SJ et al. Successful AAV8 readministration: Suppression of capsid-specific neutralizing antibodies by a combination treatment of bortezomib and CD20 mAb in a mouse model of Pompe disease. J Gene Med 2023. PMID 36994804.
- Ilyinskii PO et al. Readministration of high-dose adeno-associated virus gene therapy vectors enabled by ImmTOR nanoparticles combined with B cell-targeted agents. PNAS Nexus 2023. PMID 38024395.
- Li N et al. The Effect of Immunomodulatory Treatments on Anti-Dystrophin Immune Response After AAV Gene Therapy in Dystrophin Deficient mdx Mice. J Neuromuscul Dis 2021. PMID 34569971.
- Leborgne C et al. IgG-cleaving endopeptidase enables in vivo gene therapy in the presence of anti-AAV neutralizing antibodies. Nat Med 2020. PMID 32483358.
- Meliani A et al. Antigen-selective modulation of AAV immunogenicity with tolerogenic rapamycin nanoparticles enables successful vector re-administration. Nat Commun 2018. PMID 30291246.
- Corti M et al. Evaluation of Readministration of a Recombinant Adeno-Associated Virus Vector Expressing Acid Alpha-Glucosidase in Pompe Disease: Preclinical to Clinical Planning. Hum Gene Ther Clin Dev 2015. PMID 26390092.
- Unzu C et al. Helper-dependent adenovirus achieve more efficient and persistent liver transgene expression in non-human primates under immunosuppression. Gene Ther 2015. PMID 26125605.
- Corti M et al. B-Cell Depletion is Protective Against Anti-AAV Capsid Immune Response: A Human Subject Case Study. Mol Ther Methods Clin Dev 2014. PMID 25541616.
- Mingozzi F et al. Prevalence and pharmacological modulation of humoral immunity to AAV vectors in gene transfer to synovial tissue. Gene Ther 2013. PMID 22786533.
- Mingozzi F et al. Pharmacological modulation of humoral immunity in a nonhuman primate model of AAV gene transfer for hemophilia B. Mol Ther 2012. PMID 22565846.
- Unzu C et al. Transient and intensive pharmacological immunosuppression fails to improve AAV-based liver gene transfer in non-human primates. J Transl Med 2012. PMID 22704060.
- Fontanellas A et al. Intensive pharmacological immunosuppression allows for repetitive liver gene transfer with recombinant adenovirus in nonhuman primates. Mol Ther 2010. PMID 20087317.

F — Immune tolerance induction regimens (30)

- Kishnani PS et al. Insights into immunogenicity and therapeutic strategies to mitigate the immune response in infantile-onset Pompe disease: a comprehensive systematic literature review. Front Immunol 2025. PMID 41583481.
- Chen HA et al. Optimizing treatment outcomes: immune tolerance induction in Pompe disease patients undergoing enzyme replacement therapy. Front Immunol 2024. PMID 38715621.
- Desai AK et al. An updated management approach of Pompe disease patients with high-sustained anti-rhGAA IgG antibody titers: experience with bortezomib-based immunomodulation. Front Immunol 2024. PMID 38524130.
- Li Z et al. Eradication of FIX inhibitor in haemophilia B children using low-dose immune tolerance induction with rituximab-based immunosuppressive agent(s) in China. Haemophilia 2022. PMID 35503087.
- Tran JQ et al. Expansion of immature, nucleated red blood cells by transient low-dose methotrexate immune tolerance induction in mice. Clin Exp Immunol 2021. PMID 33205401.
- Desai AK et al. Benefits of Prophylactic Short-Course Immune Tolerance Induction in Patients With Infantile Pompe Disease: Demonstration of Long-Term Safety and Efficacy in an Expanded Cohort. Front Immunol 2020. PMID 32849613.
- Desai AK et al. The potential impact of timing of IVIG administration on the efficacy of rituximab for immune tolerance induction for patients with Pompe disease. Clin Immunol 2020. PMID 32681978.
- Gupta P et al. A Race Against Time-Changing the Natural History of CRIM Negative Infantile Pompe Disease. Front Immunol 2020. PMID 33013846.
- Julien DC et al. Immune Modulation for Enzyme Replacement Therapy in A Female Patient With Hunter Syndrome. Front Immunol 2020. PMID 32508845.
- Poelman E et al. Effects of higher and more frequent dosing of alglucosidase alfa and immunomodulation on long-term clinical outcome of classic infantile Pompe patients. J Inherit Metab Dis 2020. PMID 32506446.
- Doshi BS et al. Combined anti-CD20 and mTOR inhibition with factor VIII for immune tolerance induction in hemophilia A patients with refractory inhibitors. J Thromb Haemost 2020. PMID 31985872.

- Biswas M et al. B Cell Depletion Eliminates FVIII Memory B Cells and Enhances AAV8-coF8 Immune Tolerance Induction When Combined With Rapamycin. Front Immunol 2020. PMID 32670285.
- De Groot AS et al. HLA- and genotype-based risk assessment model to identify infantile onset pompe disease patients at high-risk of developing significant anti-drug antibodies (ADA). Clin Immunol 2019. PMID 30711607.
- Poelman E et al. High Sustained Antibody Titers in Patients with Classic Infantile Pompe Disease Following Immunomodulation at Start of Enzyme Replacement Therapy. J Pediatr 2018. PMID 29428273.
- Hassan S et al. Preventing or Eradicating Factor VIII Antibody Formation in Patients with Hemophilia A: What Can We Learn from Other Disorders? Semin Thromb Hemost 2018. PMID 30045390.
- Kazi ZB et al. Sustained immune tolerance induction in enzyme replacement therapy-treated CRIM-negative patients with infantile Pompe disease. JCI Insight 2017. PMID 28814660.
- Rairikar M et al. High dose IVIG successfully reduces high rhGAA IgG antibody titers in a CRIM-negative infantile Pompe disease patient. Mol Genet Metab 2017. PMID 28648664.
- Biswas M et al. Combination therapy for inhibitor reversal in haemophilia A using monoclonal anti-CD20 and rapamycin. Thromb Haemost 2017. PMID 27683758.
- Schultz HS et al. Quantitative analysis of the CD4+ T cell response to therapeutic antibodies in healthy donors using a novel T cell:PBMC assay. PLoS One 2017. PMID 28562666.
- Kazi ZB et al. Durable and sustained immune tolerance to ERT in Pompe disease with entrenched immune responses. JCI Insight 2016. PMID 27493997.
- Stenger EO et al. Immune Tolerance Strategies in Siblings with Infantile Pompe Disease-Advantages for a Preemptive Approach to High-Sustained Antibody Titers. Mol Genet Metab Rep 2015. PMID 26167453.
- Debiec H et al. Allo-immune membranous nephropathy and recombinant aryl sulfatase replacement therapy: a need for tolerance induction therapy. J Am Soc Nephrol 2014. PMID 24262793.
- Leissinger C et al. Rituximab for treatment of inhibitors in haemophilia A. A Phase II study. Thromb Haemost 2014. PMID 24919980.
- Banugaria SG et al. Algorithm for the early diagnosis and treatment of patients with cross reactive immunologic material-negative classic infantile pompe disease: a step towards improving the efficacy of ERT. PLoS One 2013. PMID 23825616.
- Banugaria SG et al. Bortezomib in the rapid reduction of high sustained antibody titers in disorders treated with therapeutic protein: lessons learned from Pompe disease. Genet Med 2013. PMID 23060045.
- Messinger YH et al. Successful immune tolerance induction to enzyme replacement therapy in CRIM-negative infantile Pompe disease. Genet Med 2012. PMID 22237443.
- Meslier Y et al. Bortezomib delays the onset of factor VIII inhibitors in experimental hemophilia A, but fails to eliminate established anti-factor VIII IgG-producing cells. J Thromb Haemost 2011. PMID 21251202.
- Barnes C et al. Induction of immune tolerance using rituximab in a child with severe haemophilia B with inhibitors and anaphylaxis to factor IX. Haemophilia 2010. PMID 20546030.
- Kessel C et al. Humoral immune responsiveness to a defined epitope on factor VIII before and after B cell ablation with rituximab. Mol Immunol 2008. PMID 18715645.
- Hay C et al. Current and future approaches to inhibitor management and aversion. Semin Thromb Hemost 2006. PMID 16804831.

G — Solid-tumour anti-murine-antibody precedent (14)

- Eger C et al. Generation and Characterization of a Human/Mouse Chimeric GD2-Mimicking Anti-Idiotype Antibody Ganglidiximab for Active Immunotherapy against Neuroblastoma. PLoS One 2016. PMID 26967324.
- Kushner BH et al. Prolonged progression-free survival after consolidating second or later remissions of neuroblastoma with Anti-GD2 immunotherapy and isotretinoin: a prospective Phase II study. Oncoimmunology 2015. PMID 26140243.
- Kushner BH et al. High-dose cyclophosphamide inhibition of humoral immune response to murine monoclonal antibody 3F8 in neuroblastoma patients: broad implications for immunotherapy. Pediatr Blood Cancer 2007. PMID 16421906.
- Gonzales NR et al. Minimizing immunogenicity of the SDR-grafted humanized antibody CC49 by genetic manipulation of the framework residues. Mol Immunol 2003. PMID 14522015.
- Richman CM et al. Dosimetry-based therapy in metastatic breast cancer patients using 90Y monoclonal antibody 170H.82 with autologous stem cell support and cyclosporin A. Clin Cancer Res 1999. PMID 10541370.
- Albertini MR et al. Systemic interleukin-2 modulates the anti-idiotypic response to chimeric anti-GD2 antibody in patients with melanoma. J Immunother Emphasis Tumor Immunol 1996. PMID 8877722.

- Dhingra K et al. Phase I clinical and pharmacological study of suppression of human antimouse antibody response to monoclonal antibody L6 by deoxyspergualin. Cancer Res 1995. PMID 7606728.
- Divgi CR et al. Pilot radioimmunotherapy trial with 131I-labeled murine monoclonal antibody CC49 and deoxyspergualin in metastatic colon carcinoma. Clin Cancer Res 1995. PMID 9815950.
- Kimball JA et al. The OKT3 Antibody Response Study: a multicentre study of human anti-mouse antibody (HAMA) production following OKT3 use in solid organ transplantation. Transpl Immunol 1995. PMID 8581409.
- Weiden PL et al. Human anti-mouse antibody suppression with cyclosporin A. Cancer 1994. PMID 8306252.
- Rettenbacher L et al. [Anaphylactic shock after repeated injection of 99mTc-labeled CEA antibody]. Nuklearmedizin 1994. PMID 8090628.
- Hammond EA et al. Prevention of adverse clinical outcome by monitoring of cardiac transplant patients for murine monoclonal CD3 antibody (OKT3) sensitization. Transplantation 1993. PMID 8497882.
- Tjandra JJ et al. Development of human anti-murine antibody (HAMA) response in patients. Immunol Cell Biol 1990. PMID 1711007.
- Ledermann JA et al. Repeated antitumour antibody therapy in man with suppression of the host response by cyclosporin A. Br J Cancer 1988. PMID 3265333.

H — Lymphodepletion and concomitant immunosuppression (8)

- Christodoulou I et al. Comparative Analysis of Lymphodepletion Regimens in CART-19 Treatment for Relapsed/Refractory Diffuse Large B Cell Lymphoma. Transplant Cell Ther 2026. PMID 41183747.
- Synnott D et al. Systematic Review of Immunosuppression After Chimeric Antigen Receptor T-Cell Therapy for Posttransplant Lymphoproliferative Disorder. Kidney Int Rep 2026. PMID 41907816.
- Ozog S et al. Influence of B cell-lineage targeted CAR-T cell therapy on humoral immunity and vaccine-induced antibody response. Nat Commun 2026. PMID 41942445.
- Roddie C et al. Matched donor allogeneic CAR-T for adult B-ALL: toxicity, efficacy, repeat dosing, and the importance of lymphodepletion. Blood 2025. PMID 40643148.
- Aleissa MM et al. Severe Acute Respiratory Syndrome Coronavirus 2 Vaccine Immunogenicity among Chimeric Antigen Receptor T Cell Therapy Recipients. Transplant Cell Ther 2023. PMID 36906276.
- Iqbal M et al. Impact of Rituximab and Corticosteroids on Late Cytopenias Post-Chimeric Antigen Receptor T Cell Therapy. Transplant Cell Ther 2022. PMID 35842124.
- Walti CS et al. Humoral immunogenicity of the seasonal influenza vaccine before and after CAR-T-cell therapy: a prospective observational study. J Immunother Cancer 2021. PMID 34702753.
- Graalmann T et al. B cell depletion impairs vaccination-induced CD8+ T cell responses in a type I interferon-dependent manner. Ann Rheum Dis 2021. PMID 34226189.

I — Confusables (not evidence) (9)

- BrainChild Bio, Inc (sponsor). Illuminate: a phase 2 pivotal study of BCB-276, a B7-H3-specific CAR T cell locoregional immunotherapy for DIPG - NCT07680439. ClinicalTrials.gov 2026. https://clinicaltrials.gov/study/NCT07680439.
- Li Y et al. Rituximab potentially improves clinical outcomes of CAR-T therapy for r/r B-ALL via sensitizing leukemia cells to CAR-T-mediated cytotoxicity and reducing CAR-T exhaustion. Cell Oncol (Dordr) 2024. PMID 38662336.
- Xiong X et al. Functional Validation of the RQR8 Suicide /Marker Gene in CD19 CAR-T Cells and CLL1CAR-T Cells. Ann Hematol 2023. PMID 37086278.
- Mosti L et al. Targeted multi-epitope switching enables straightforward positive/negative selection of CAR T cells. Gene Ther 2021. PMID 33526841.
- Sang W et al. Phase II trial of co-administration of CD19- and CD20-targeted chimeric antigen receptor T cells for relapsed and refractory diffuse large B cell lymphoma. Cancer Med 2020. PMID 32608579.
- Viaud S et al. Switchable control over in vivo CAR T expansion, B cell depletion, and induction of memory. Proc Natl Acad Sci U S A 2018. PMID 30373813.
- Philip B et al. A highly compact epitope-based marker/suicide gene for easier and safer T-cell therapy. Blood 2014. PMID 24970931.
- van Loenen MM et al. Multi-cistronic vector encoding optimized safety switch for adoptive therapy with T-cell receptor-modified T cells. Gene Ther 2013. PMID 23364317.
- Griffioen M et al. Retroviral transfer of human CD20 as a suicide gene for adoptive T-cell therapy. Haematologica 2009. PMID 19734426.

J — Premedication and infusion reactions (15)

- Nguyen NN et al. Comparison of H1 versus combined H1/H2 antagonist premedication on incidence of rituximab infusion-related reaction: A retrospective cohort study Journal of oncology pharmacy practice : official publication of the International Society of Oncology Pharmacy Practitioners 2026. PMID 42530421.
- El Shenawy Z et al. Reevaluating corticosteroid premedication for rituximab: a retrospective cohort study Leukemia & lymphoma 2026. PMID 41789935.
- Bovens P et al. Oral glucocorticoid premedication for preventing infusion-related reactions to rituximab in rheumatoid arthritis Scandinavian journal of rheumatology 2026. PMID 41572843.
- Kitahiro Y et al. Efficacy of bepotastine compared with hydroxyzine in preventing rituximab-induced infusion-related reactions in non-hodgkin lymphoma patients: a phase II, double-blind, multicenter, and randomized trial International journal of clinical oncology 2026. PMID 41989666.
- Candar Ö et al. Management of Rituximab-Associated Hypersensitivity Reactions with Successfully Applied Desensitization Protocols: A Clinical Experience of 46 Infusions in 11 Patients Journal of clinical medicine 2026. PMID 42279029.
- Feltesse C et al. Severe Reactions to Rituximab in Children: A Cohort Study of Rituximab-Induced Serum Sickness and Anaphylaxis Children (Basel, Switzerland) 2026. PMID 42073020.
- Mori F et al. Immediate and delayed hypersensitivity to biologicals in children-A practical approach: An EAACI task force report Pediatric allergy and immunology : official publication of the European Society of Pediatric Allergy and Immunology 2026. PMID 41937341.
- Barroso A et al. Management of infusion-related reactions in cancer therapy: strategies and challenges ESMO open 2024. PMID 38452439.
- Karlsen EA et al. Steroid Premedication and Monoclonal Antibody Therapy: Should We Reconsider? Current treatment options in oncology 2024. PMID 38270799.
- Ding J et al. A novel prednisone premedication protocol significantly decreases infusion-related reactions of rituximab in newly diagnosed diffuse large B-cell lymphoma Oncology letters 2023. PMID 37205922.
- Cáceres MC et al. The importance of early identification of infusion-related reactions to monoclonal antibodies Therapeutics and clinical risk management 2019. PMID 31447561.
- Vogel WH. Infusion reactions: diagnosis, assessment, and management Clinical journal of oncology nursing 2010. PMID 20350882.
- Tuthill M et al. Rapid infusion of rituximab over 60 min European journal of haematology 2009. PMID 19220420.
- Sehn LH et al. Rapid infusion rituximab in combination with corticosteroid-containing chemotherapy or as maintenance therapy is well tolerated and can safely be delivered in the community setting Blood 2007. PMID 17244675.
- Salar A et al. Rapid infusion of rituximab with or without steroid-containing chemotherapy: 1-yr experience in a single institution European journal of haematology 2006. PMID 16856919.

K — Screening, prophylaxis, and late effects (21)

- Lee E et al. Infectious complications after CAR T-cell therapy: mechanisms, risk stratification, and prevention Blood research 2026. PMID 42274870.
- Kamboj M et al. Vaccination of Adults With Cancer: ASCO Guideline Journal of clinical oncology : official journal of the American Society of Clinical Oncology 2024. PMID 38498792.
- Athni TS et al. Hypogammaglobulinemia, late-onset neutropenia, and infections following rituximab Annals of allergy, asthma & immunology : official publication of the American College of Allergy, Asthma, & Immunology 2023. PMID 36706910.
- Wat J et al. Hypogammaglobulinemia After Chimeric Antigen Receptor (CAR) T-Cell Therapy: Characteristics, Management, and Future Directions The journal of allergy and clinical immunology. In practice 2022. PMID 34757064.
- Hayden PJ et al. Management of adults and children receiving CAR T-cell therapy: 2021 best practice recommendations of the European Society for Blood and Marrow Transplantation (EBMT) and the Joint Accreditation Committee of ISCT and EBMT (JACIE) and the European Haematology Association (EHA) Annals of oncology : official journal of the European Society for Medical Oncology 2022. PMID 34923107.
- Moor MB et al. Humoral and cellular responses to mRNA vaccines against SARS-CoV-2 in patients with a history of CD20 B-cell-depleting therapy (RituxiVac): an investigator-initiated, single-centre, open-label study The Lancet. Rheumatology 2021. PMID 34514436.

- Walti CS et al. Antibodies against vaccine-preventable infections after CAR-T cell therapy for B cell malignancies JCI insight 2021. PMID 33914708.
- Hwang JP et al. Hepatitis B Virus Screening and Management for Patients With Cancer Prior to Therapy: ASCO Provisional Clinical Opinion Update Journal of clinical oncology : official journal of the American Society of Clinical Oncology 2020. PMID 32716741.
- Bar-Or A et al. Effect of ocrelizumab on vaccine responses in patients with multiple sclerosis: The VELOCE study Neurology 2020. PMID 32727835.
- Yakoub-Agha I et al. Management of adults and children undergoing chimeric antigen receptor T-cell therapy: best practice recommendations of the European Society for Blood and Marrow Transplantation (EBMT) and the Joint Accreditation Committee of ISCT and EBMT (JACIE) Haematologica 2020. PMID 31753925.
- Hill JA et al. CAR-T - and a side order of IgG, to go? - Immunoglobulin replacement in patients receiving CAR-T cell therapy Blood reviews 2019. PMID 31416717.
- Barmettler S et al. Association of Immunoglobulin Levels, Infectious Risk, and Mortality With Rituximab and Hypogammaglobulinemia JAMA network open 2018. PMID 30646343.
- Loomba R et al. Hepatitis B Reactivation Associated With Immune Suppressive and Biological Modifier Therapies: Current Concepts, Management Strategies, and Future Directions Gastroenterology 2017. PMID 28219691.
- Hwang JP et al. Hepatitis B Virus Screening for Patients With Cancer Before Therapy: American Society of Clinical Oncology Provisional Clinical Opinion Update Journal of clinical oncology : official journal of the American Society of Clinical Oncology 2015. PMID 25964247.
- Huang YH et al. Randomized controlled trial of entecavir prophylaxis for rituximab-associated hepatitis B virus reactivation in patients with lymphoma and resolved hepatitis B Journal of clinical oncology : official journal of the American Society of Clinical Oncology 2013. PMID 23775967.
- Casulo C et al. Incidence of hypogammaglobulinemia in patients receiving rituximab and the use of intravenous immunoglobulin for recurrent infections Clinical lymphoma, myeloma & leukemia 2013. PMID 23276889.
- Tesfa D et al. Late-onset neutropenia following rituximab therapy in rheumatic diseases: association with B lymphocyte depletion and infections Arthritis and rheumatism 2011. PMID 21560117.
- Howard SC et al. The tumor lysis syndrome The New England journal of medicine 2011. PMID 21561350.
- Wolach O et al. Late-onset neutropenia after rituximab treatment: case series and comprehensive review of the literature Medicine 2010. PMID 20827108.
- Yeo W et al. Hepatitis B virus reactivation in lymphoma patients with prior resolved hepatitis B undergoing anticancer therapy with or without rituximab Journal of clinical oncology : official journal of the American Society of Clinical Oncology 2009. PMID 19075267.
- Carson KR et al. Progressive multifocal leukoencephalopathy after rituximab therapy in HIV-negative patients: a report of 57 cases from the Research on Adverse Drug Events and Reports project Blood 2009. PMID 19264918.

L — Dose, route, and pharmacokinetics (9)

- Hartinger JM et al. Prevalence of anti-rituximab antibodies in autoimmune forms of glomerulonephritis and podocytopathies Nephrology, dialysis, transplantation : official publication of the European Dialysis and Transplant Association - European Renal Association 2026. PMID 42340364.
- Wang D et al. The impact of anti-rituximab antibodies on rituximab efficacy in children with frequently relapsing or steroid-dependent nephrotic syndrome Pediatric nephrology (Berlin, Germany) 2026. PMID 42481732.
- Harirchian MH et al. The Sanctuary Within: Development of CD20+ CNS Lymphoma Despite Peripheral B-Cell Depletion by Rituximab in a Multiple Sclerosis Patient Case reports in neurological medicine 2026. PMID 42211720.
- Lee GY et al. Retrospective Review of Intra-Cerebrospinal Fluid (CSF) Drug Delivery in CNS Malignancies: Safety, Clinical Efficacy and Pharmacokinetic Profiles of Intracerebroventricular (ICV), Lumbar Intrathecal (LIT), and Intra-Cisterna Magna (ICM) Injections Cancers 2025. PMID 40282439.
- Hartinger JM et al. Implications of rituximab pharmacokinetic and pharmacodynamic alterations in various immune-mediated glomerulopathies and potential anti-CD20 therapy alternatives Frontiers in immunology 2022. PMID 36420256.
- Locke KW et al. ENHANZE® drug delivery technology: a novel approach to subcutaneous administration using recombinant human hyaluronidase PH20 Drug delivery 2019. PMID 30744432.
- Cohen HP et al. Switching Reference Medicines to Biosimilars: A Systematic Literature Review of Clinical Outcomes Drugs 2018. PMID 29500555.
- Ryman JT et al. Pharmacokinetics of Monoclonal Antibodies CPT: pharmacometrics & systems pharmacology 2017. PMID 28653357.

- Shpilberg O et al. Subcutaneous administration of rituximab (MabThera) and trastuzumab (Herceptin) using hyaluronidase British journal of cancer 2013. PMID 24002601.

M — CAR-T round context (10)

- Ronsley R et al. Tumor inflammation-associated neurotoxicity in children with diffuse intrinsic pontine glioma receiving B7-H3-targeting CAR T cells on BrainChild-03 Neuro-oncology practice 2026. PMID 41798119.
- Ramsoomair CK et al. Locoregional delivery of CAR T cells in high-grade gliomas: a systematic analysis of safety, efficacy, and emerging biomarkers of response Journal for immunotherapy of cancer 2026. PMID 41895715.
- Canelo-Vilaseca M et al. Lymphodepletion chemotherapy in chimeric antigen receptor-engineered T (CAR-T) cell therapy in lymphoma Bone marrow transplantation 2025. PMID 40148484.
- Varela-González-Aller J et al. Towards Personalized Lymphodepletion: A Population Pharmacokinetic Fludarabine Model in Patients Receiving CAR T-Cell Therapy Pharmaceutics 2025. PMID 41471106.
- Panetta JC et al. Age-adjusted dosing of fludarabine for lymphodepletion in CAR T-cell therapy: a clinical trial simulation study Blood advances 2025. PMID 40609087.
- Wicha SG et al. Chimeric antigen receptor T-cell therapy and fludarabine: precision dosing imperatives Blood advances 2024. PMID 38191740.
- Bagley SJ et al. Intrathecal bivalent CAR T cells targeting EGFR and IL13Rα2 in recurrent glioblastoma: phase 1 trial interim results Nature medicine 2024. PMID 38480922.
- Vitanza NA et al. Intraventricular B7-H3 CAR T Cells for Diffuse Intrinsic Pontine Glioma: Preliminary First-in-Human Bioactivity and Safety Cancer discovery 2023. PMID 36259971.
- Donovan LK et al. Locoregional delivery of CAR T cells to the cerebrospinal fluid for treatment of metastatic medulloblastoma and ependymoma Nature medicine 2020. PMID 32341580.
- Lee DW et al. ASTCT Consensus Grading for Cytokine Release Syndrome and Neurologic Toxicity Associated with Immune Effector Cells Biology of blood and marrow transplantation : journal of the American Society for Blood and Marrow Transplantation 2019. PMID 30592986.

N — Procedural fasting guidelines (4)

- Joshi GP et al. 2023 American Society of Anesthesiologists Practice Guidelines for Preoperative Fasting: Carbohydrate-containing Clear Liquids with or without Protein, Chewing Gum, and Pediatric Fasting Duration-A Modular Update of the 2017 American Society of Anesthesiologists Practice Guidelines for Preoperative Fasting Anesthesiology 2023. PMID 36629465.
- Frykholm P et al. Pre-operative fasting in children: A guideline from the European Society of Anaesthesiology and Intensive Care European journal of anaesthesiology 2022. PMID 34857683.
- ASA Task Force on Preoperative Fasting. Practice Guidelines for Preoperative Fasting and the Use of Pharmacologic Agents to Reduce the Risk of Pulmonary Aspiration: Application to Healthy Patients Undergoing Elective Procedures: An Updated Report by the American Society of Anesthesiologists Task Force on Preoperative Fasting and the Use of Pharmacologic Agents to Reduce the Risk of Pulmonary Aspiration Anesthesiology 2017. PMID 28045707.
- Smith I et al. Perioperative fasting in adults and children: guidelines from the European Society of Anaesthesiology European journal of anaesthesiology 2011. PMID 21712716.